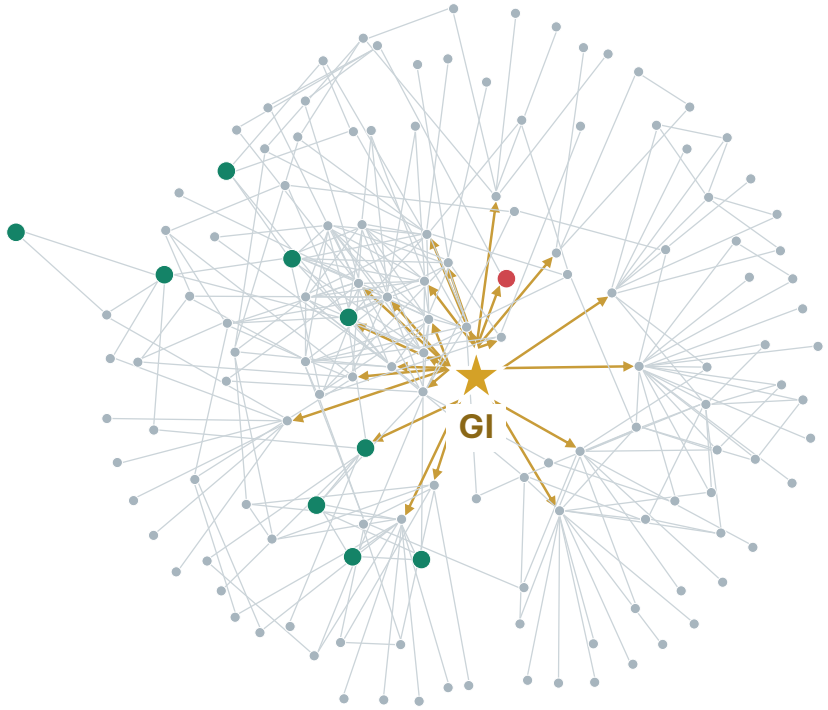
Cornell
SC Johnson College of Business

CornellBowers
College of Computing + Information Science

Search Incursions

A Relational Theory of Disequilibrating Inventions and Recombinant Advantage



Tejas Ramdas

Cornell University
Joint work with Gautam Ahuja

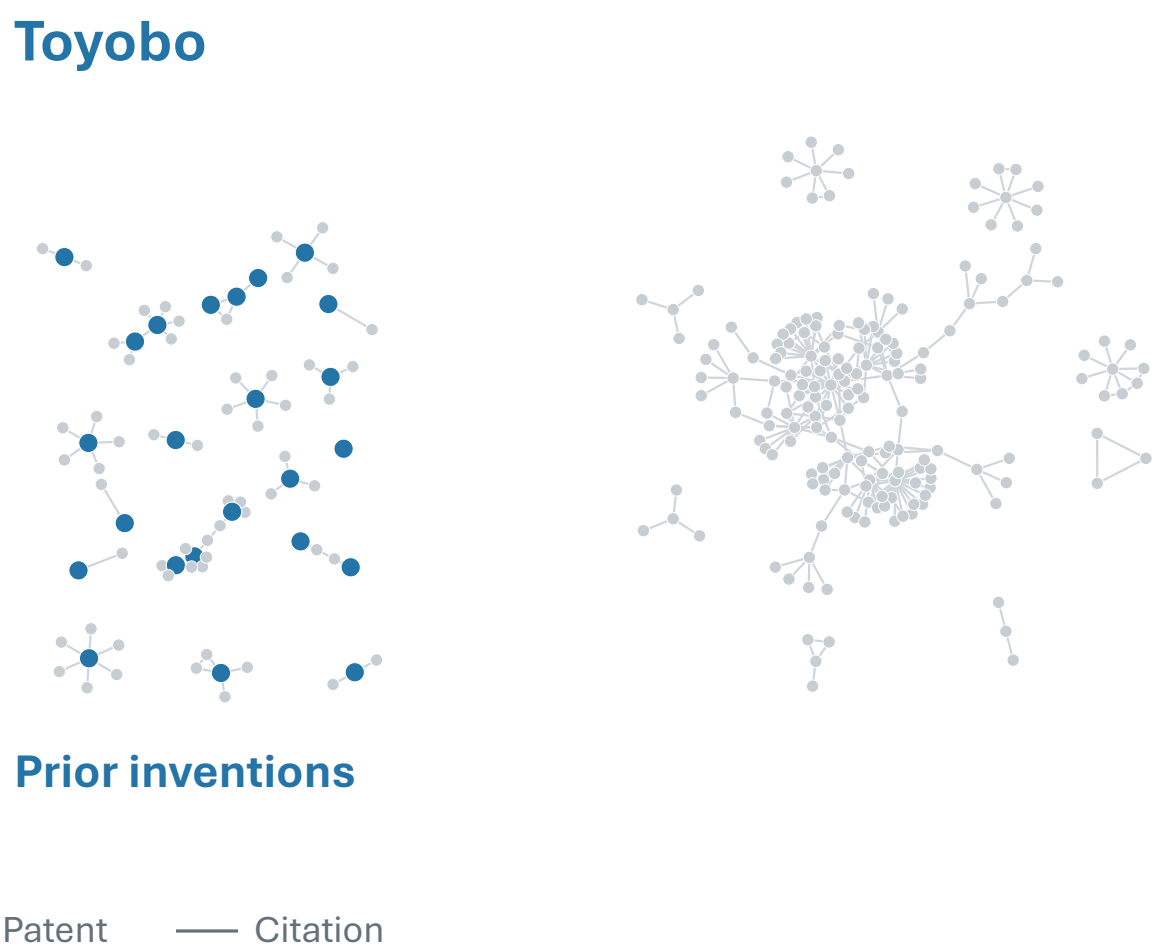
Presented at: Texas McCombs
November 11, 2026

Motivation

- **Toyobo** traces its origins to a **cotton-spinning enterprise** founded in Japan in **1882**.

- By the 1980s, Toyobo manufactured **synthetic fibers, polymer films, plastics, and industrial materials**.

Toyobo's prior inventions

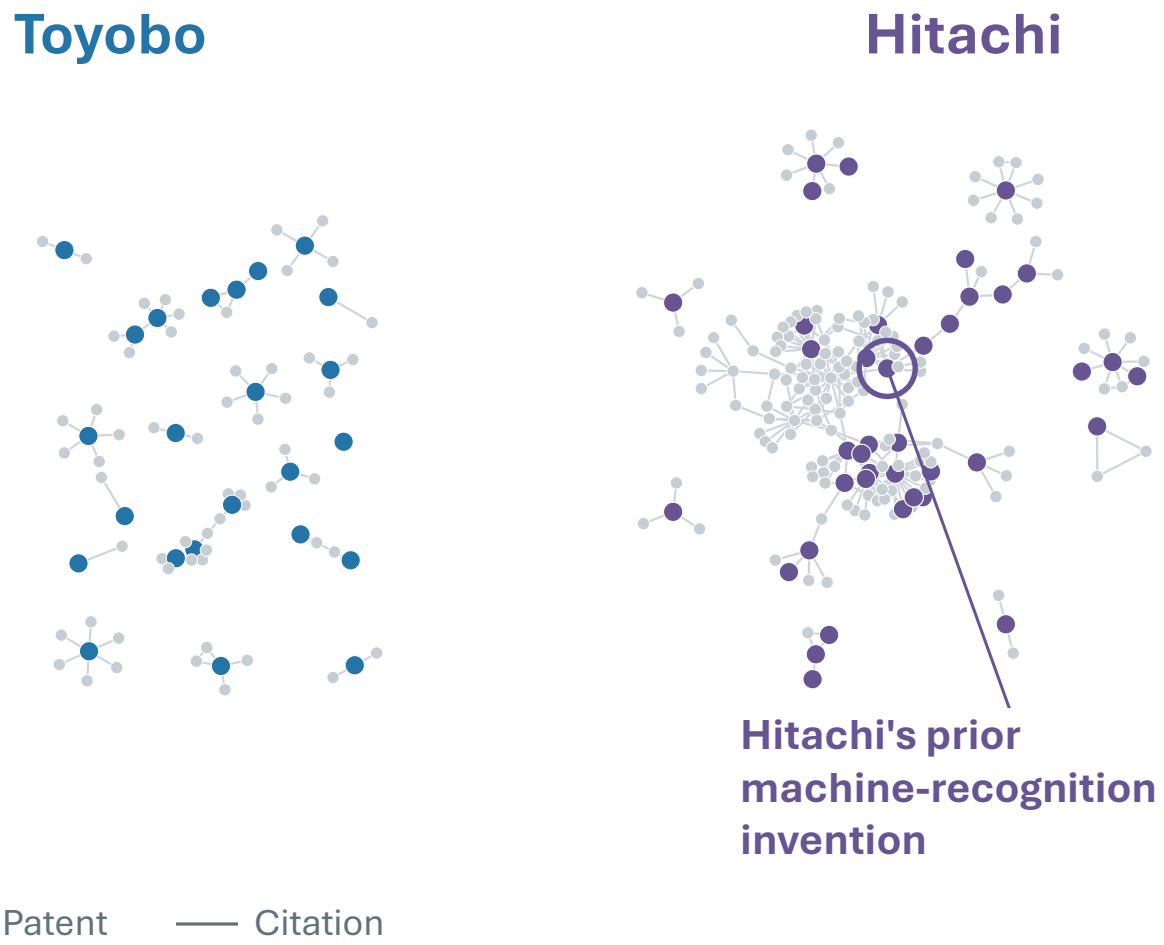


Motivation

- **Hitachi** began in **1910** as a repair shop for mining machinery.

- By the 1980s, its technologies included inventions enabling machines to **recognize the shapes and positions of objects**.

Firms and their prior inventions

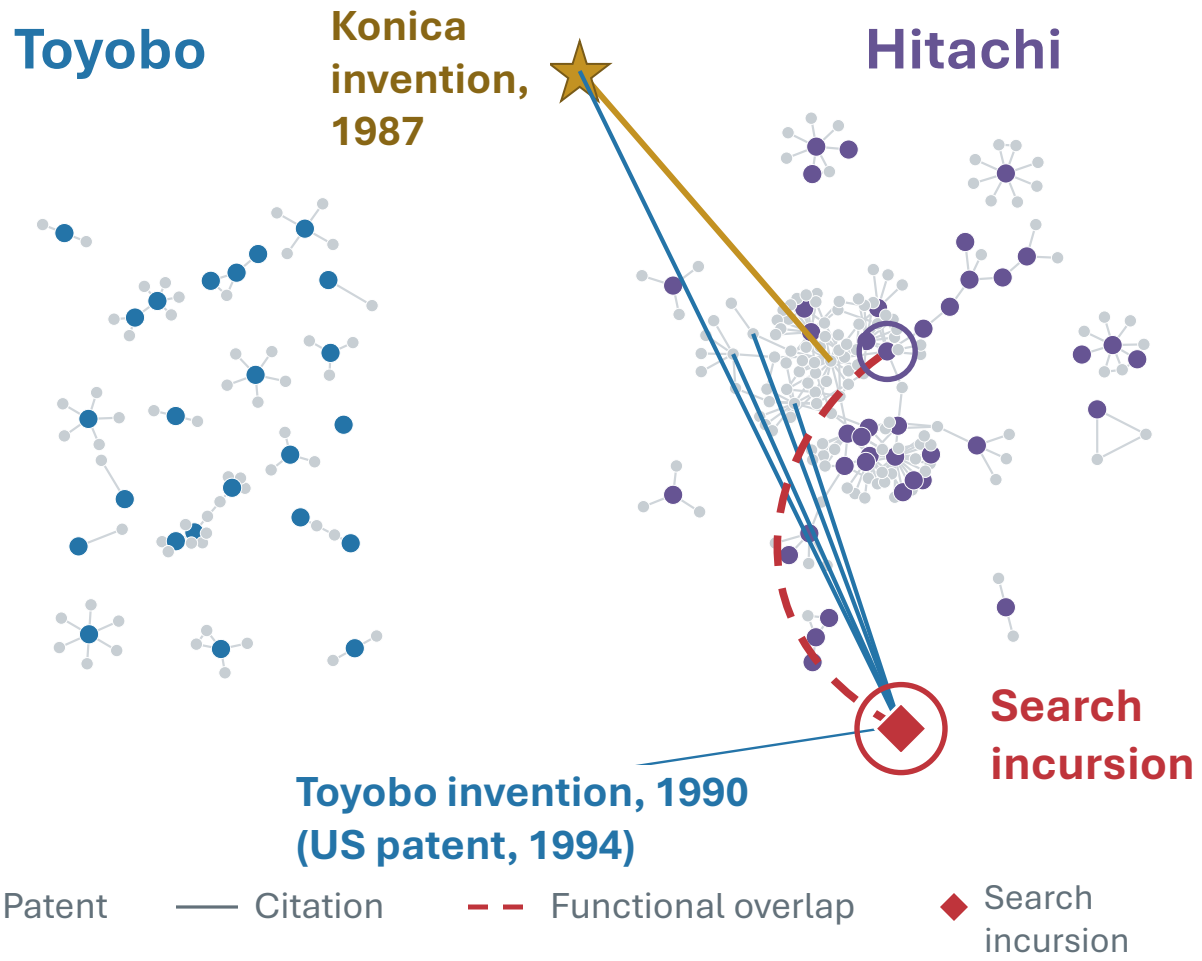


Motivation

- In **1987**, **Konica** introduced an invention that improved the conversion of visual information into **computer-readable signals while preserving important details.**

- By **1990**, **Toyobo** appears to have built upon Konica's invention and its own technological knowledge to develop a **novel method for preserving shape information in digital images**, entering an inventive region previously associated with **Hitachi**.

An invention and a search incursion

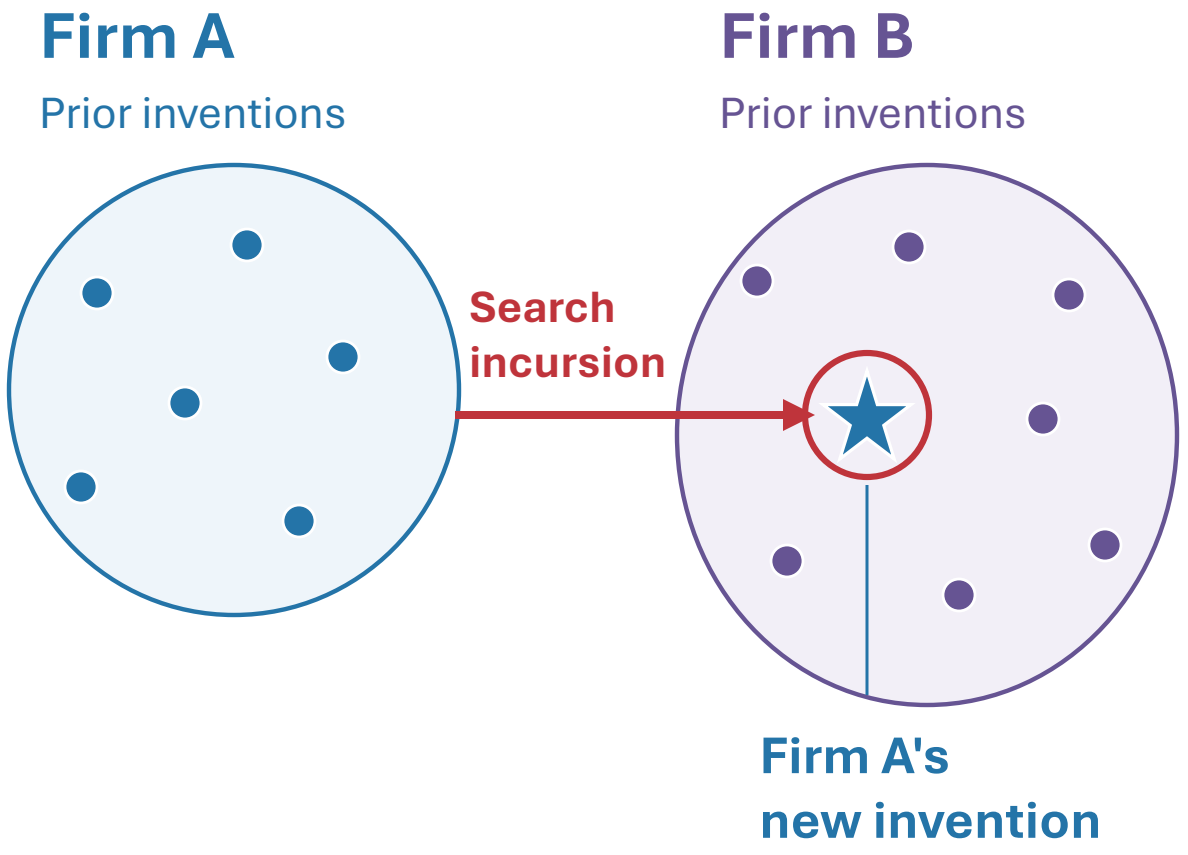


Search Incursion

A **search incursion** occurs when a firm develops a novel invention:

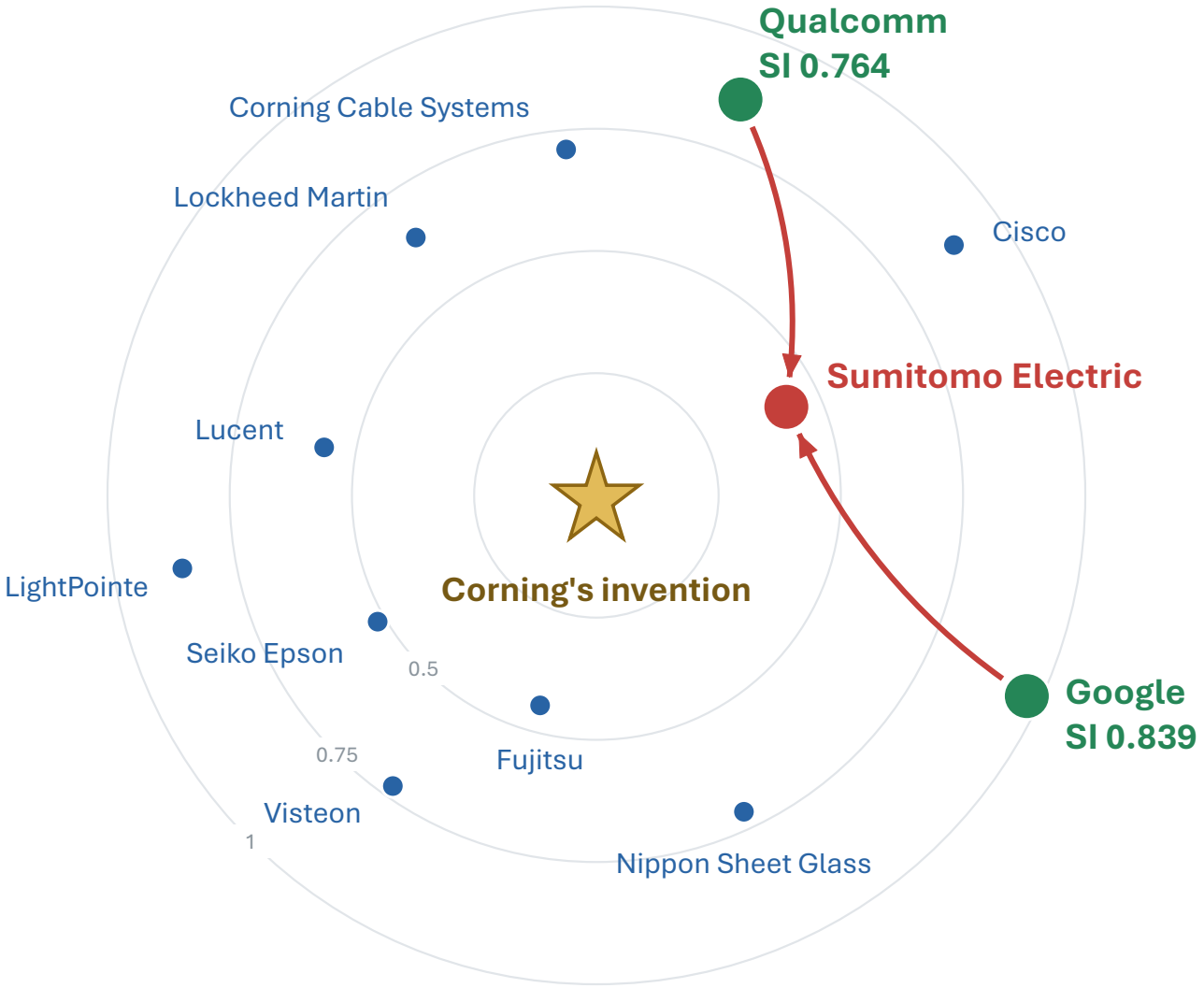
- **Outside its own established inventive region**, and
- **Within a region associated with another firm's prior inventive activity.**

A search incursion



Optical-Fiber Alignment

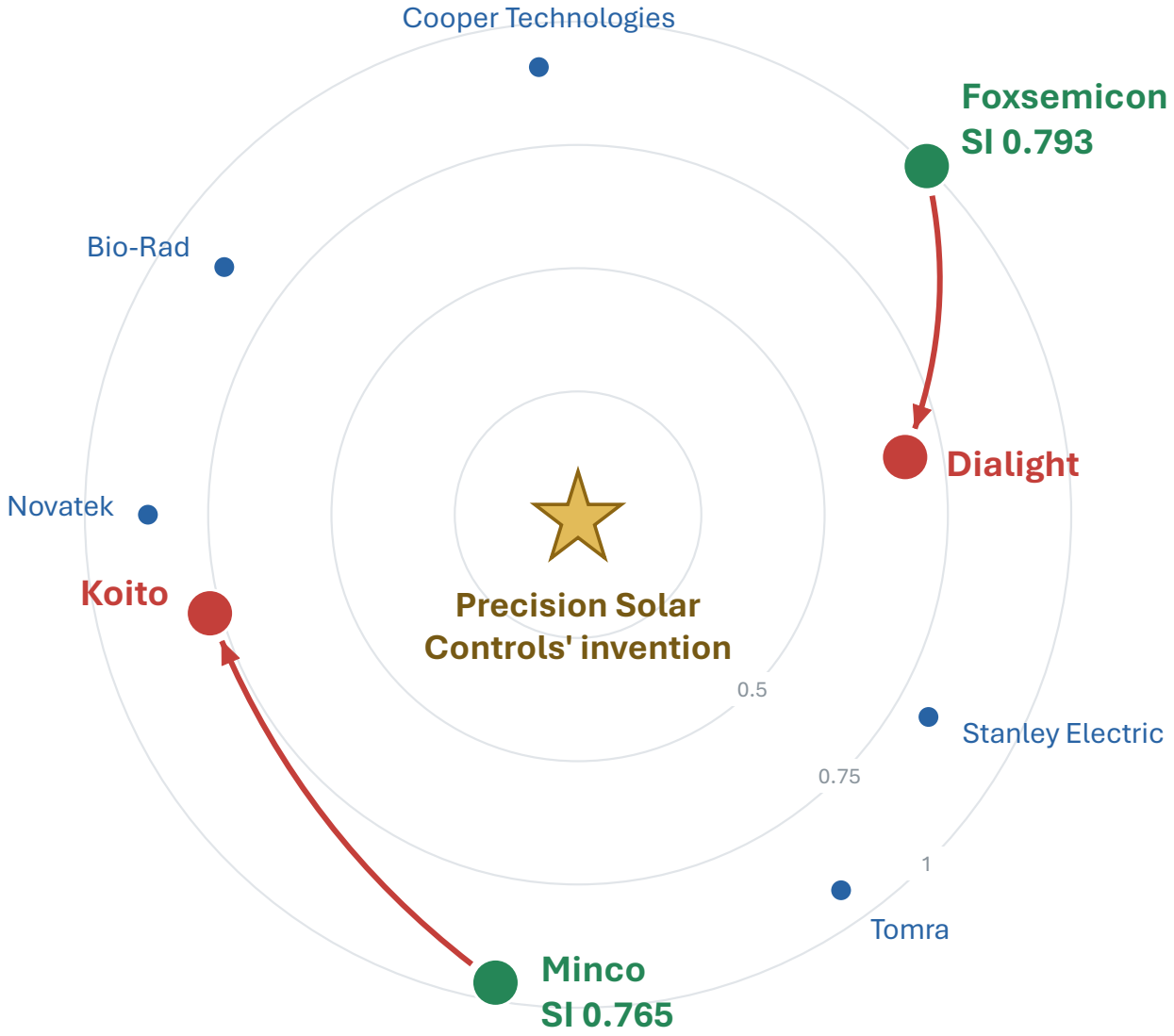
Corning | 2003 | US 6,587,618



Radius: technological distance | 12 measured firms | SI cutoff: 0.709

De-icing System for Traffic Signals

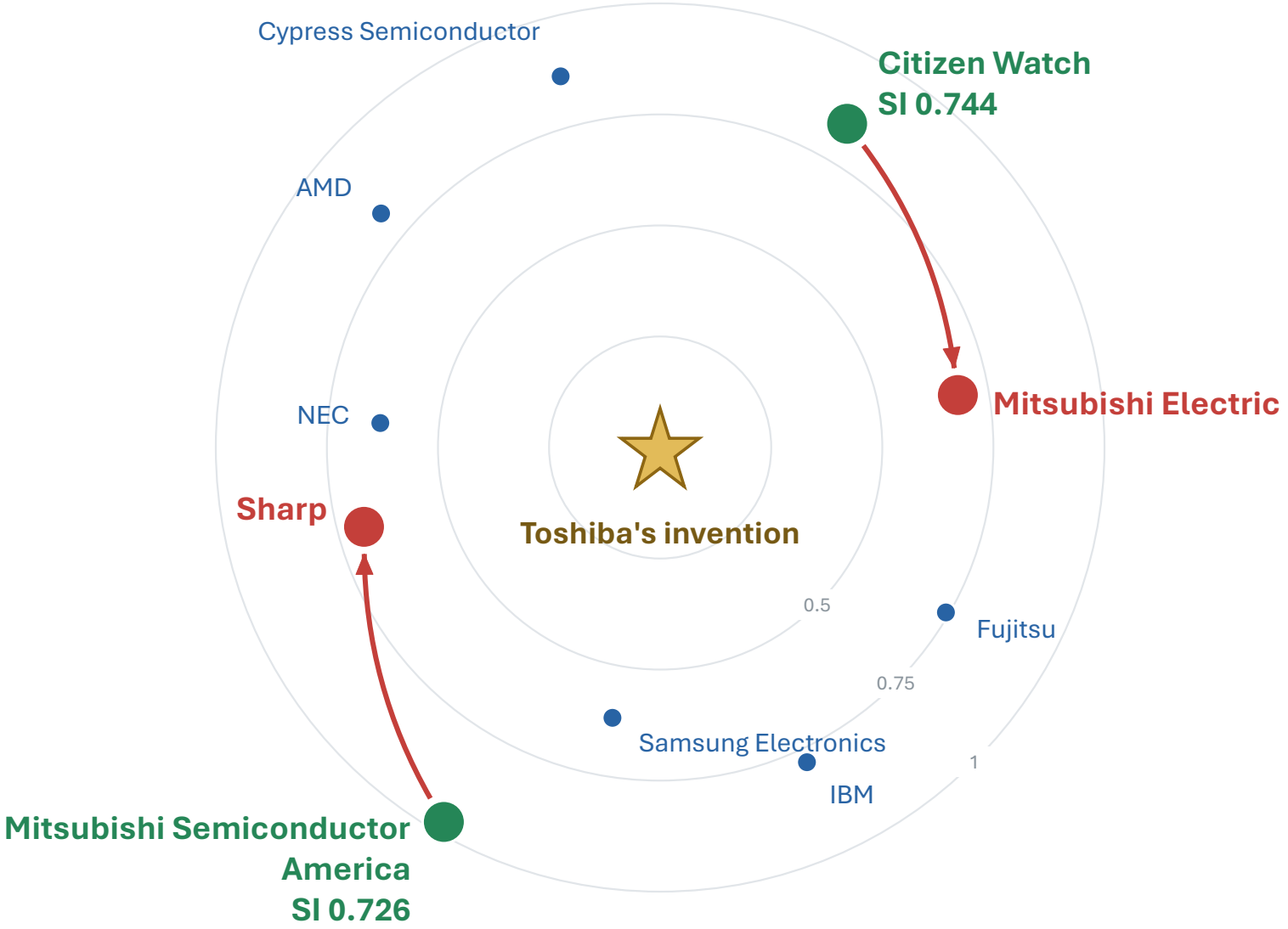
Precision Solar Controls | 2007 | US 7,211,771



Radius: technological distance | 9 measured firms | SI cutoff: 0.709

Nonvolatile Semiconductor Memory

Toshiba | 1995 | US 5,438,542

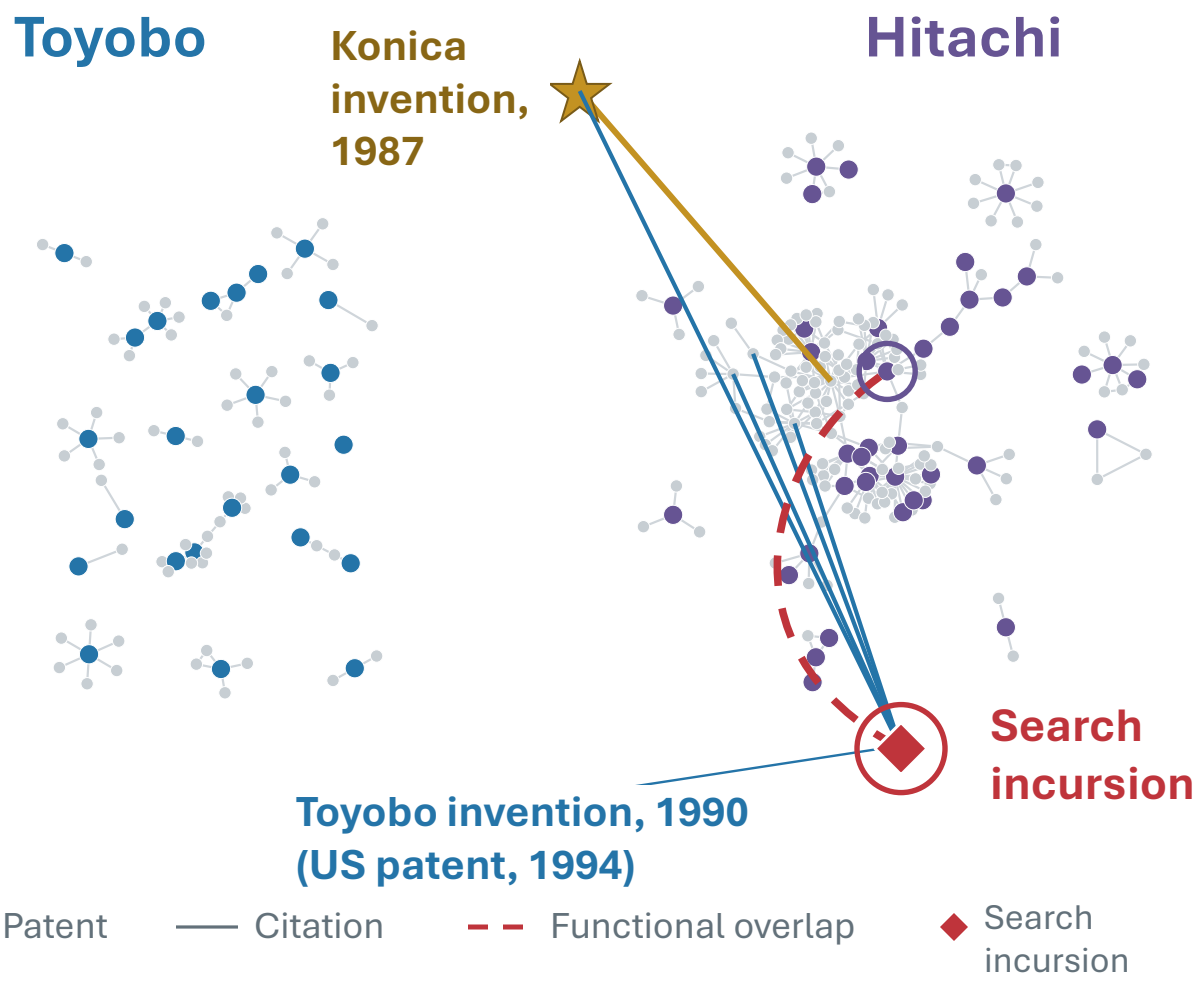


Radius: technological distance | 10 measured firms | SI cutoff: 0.709

Research Questions

RQ1. How might **Konica's invention** have enabled **Toyobo** to invent in **Hitachi's established inventive region**?

An invention and a search incursion

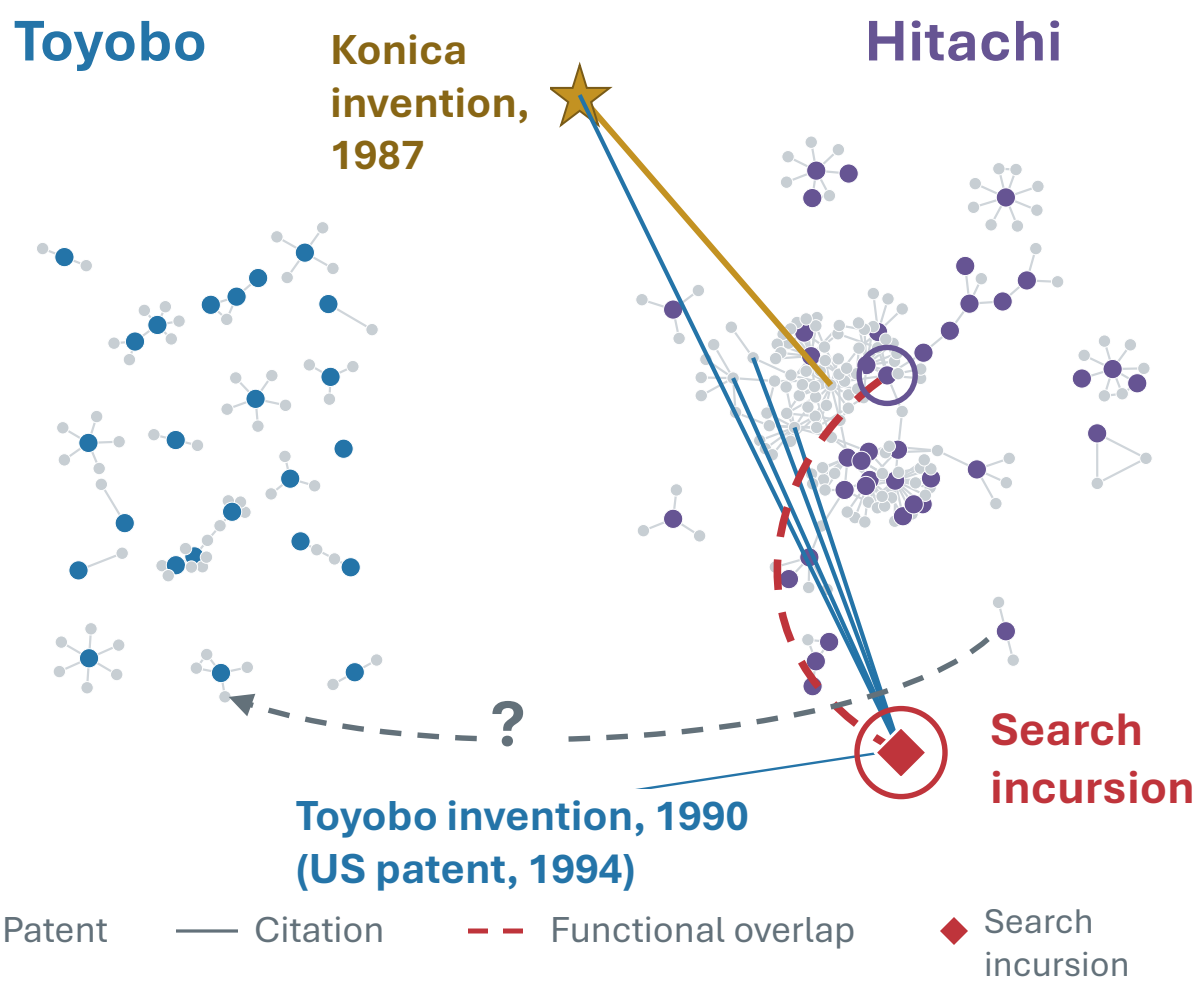


Research Questions

RQ1. How might **Konica's invention** have enabled **Toyobo** to invent in **Hitachi's established inventive region**?

RQ2. Why did **Toyobo enter Hitachi's inventive region**, rather than **Hitachi entering Toyobo's**?

An invention and a search incursion



Research Questions

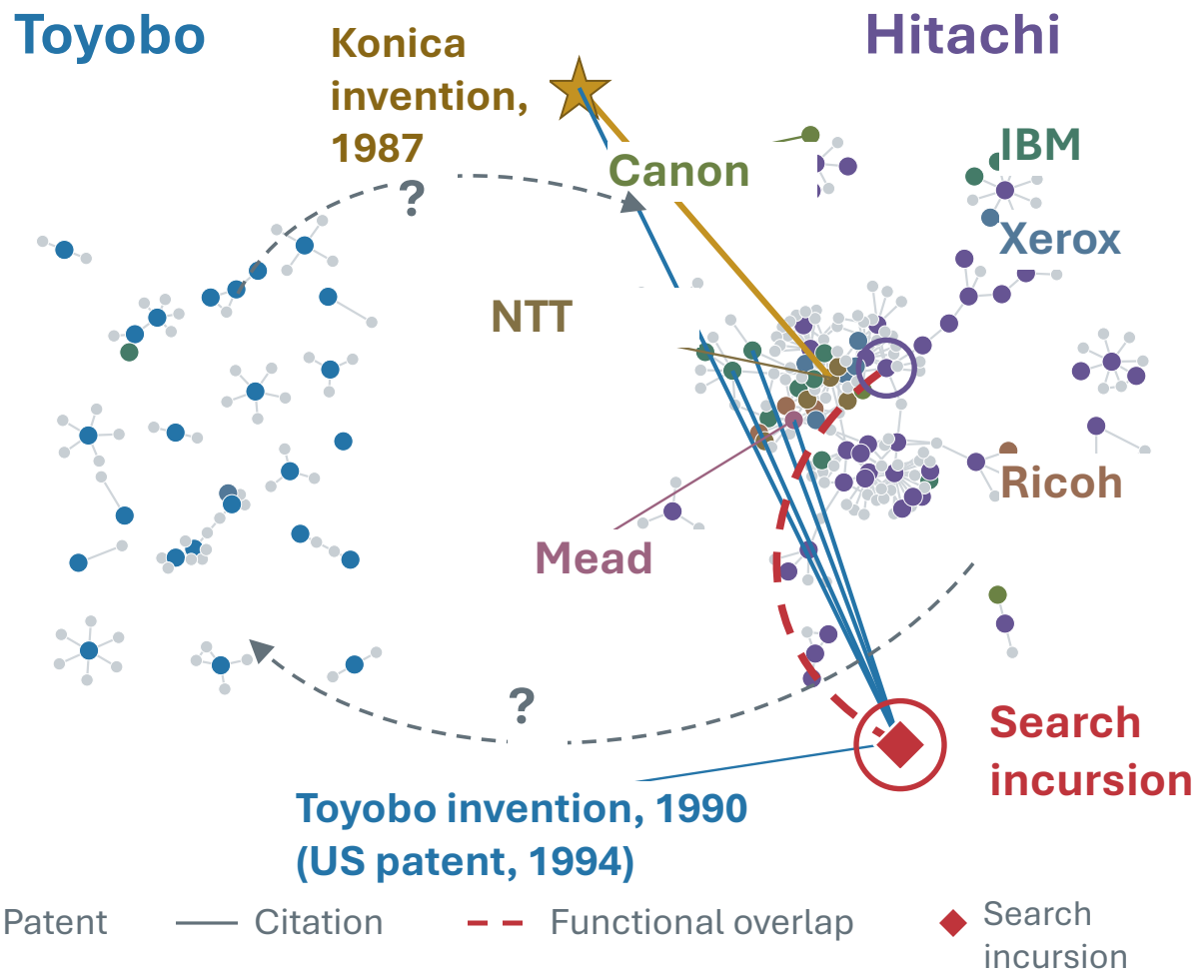
RQ1. How might **Konica's invention** have enabled **Toyobo** to invent in **Hitachi's established inventive region**?

RQ2. Why did **Toyobo enter Hitachi's inventive region**, rather than **Hitachi entering Toyobo's**?

RQ3. For **each invention**, can we explain:

- **Which particular firms** gain opportunities to undertake search incursions?
- **Which particular firms** become vulnerable to incursions?
- **Which particular firms** enter **which other firms' established inventive regions?**

An invention and a search incursion



Prior Research

- **Innovation search:** How search practices, knowledge boundaries, and search costs shape inventive activity
  (Ahuja and Lampert, 2001; Rosenkopf and Nerkar, 2001; Kwon et al., 2025).

- **Recombinant innovation:** How knowledge interdependencies and new technological bridges enable recombination
  (Fleming and Sorenson, 2001; Qu, Eggers, and Kumar, 2026).

- **Knowledge networks and technological positioning:** How knowledge structures and firms' technological positions shape inventive possibilities
  (Stuart and Podolny, 1996; Carnabuci and Operti, 2013; Arts, Cassiman, and Hou, 2023).

- **Technological change:** How discontinuities reshape competitive advantage and firms' responses
  (Tushman and Anderson, 1986; Tripsas, 1997; Li, 2024).

Roadmap

1. **Why Study Search Incursions?**
2. **Prior Literature**
3. **Theory and Hypotheses**
4. **Data and Empirical Analysis**
5. **Discussion and Conclusions**

Why Study Search Incursions?

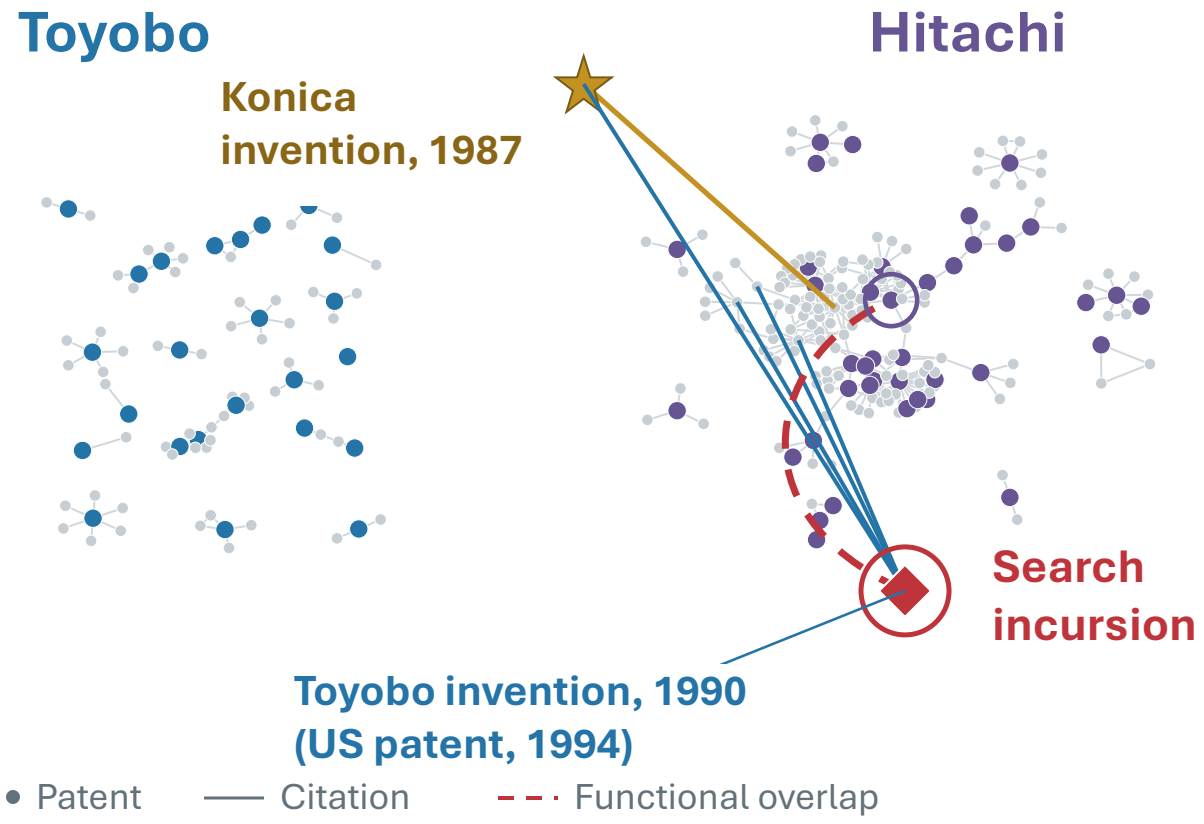
1. Consequences for firms' competitive positions

- Search incursions may introduce **competing intellectual property claims**, constraining firms' ability to appropriate value from prior inventions

  (Ziedonis, 2004).

- Even without intellectual property conflicts, incursions may constrain **strategic autonomy**, shaping commercialization and subsequent inventive choices

  (Stuart and Podolny, 1996; Katila and Chen, 2008).



Potential consequences

Appropriability **Strategic autonomy**

Why Study Search Incursions?

2. Understanding broader technological transformations

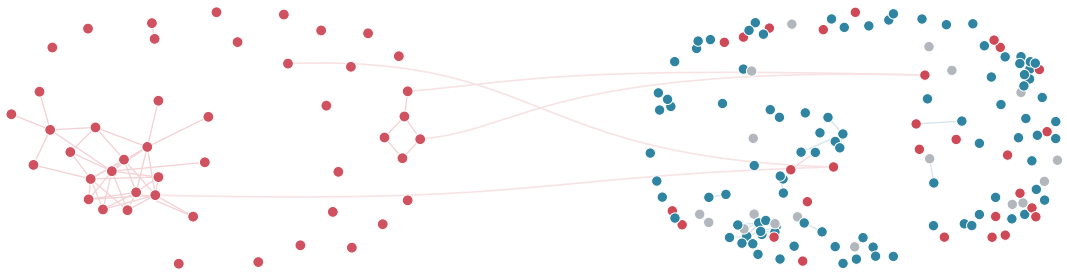
- Research on **technological convergence** (Rosenberg, 1963; Uzunca, 2018) and **competency-destroying discontinuities** (Tushman and Anderson, 1986; Rothaermel and Hill, 2005) examines broad shifts in inventive competition.

● Previously focused elsewhere ● Initial patenting leaders ● Other / unknown

1999

Firms and their earlier patents in other classes

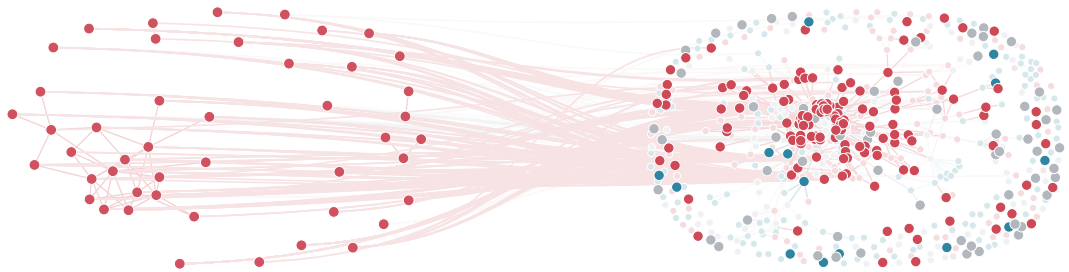
Firms and their patents in electrolytic material removal



2009

Firms and their earlier patents in other classes

Firms and their patents in electrolytic material removal



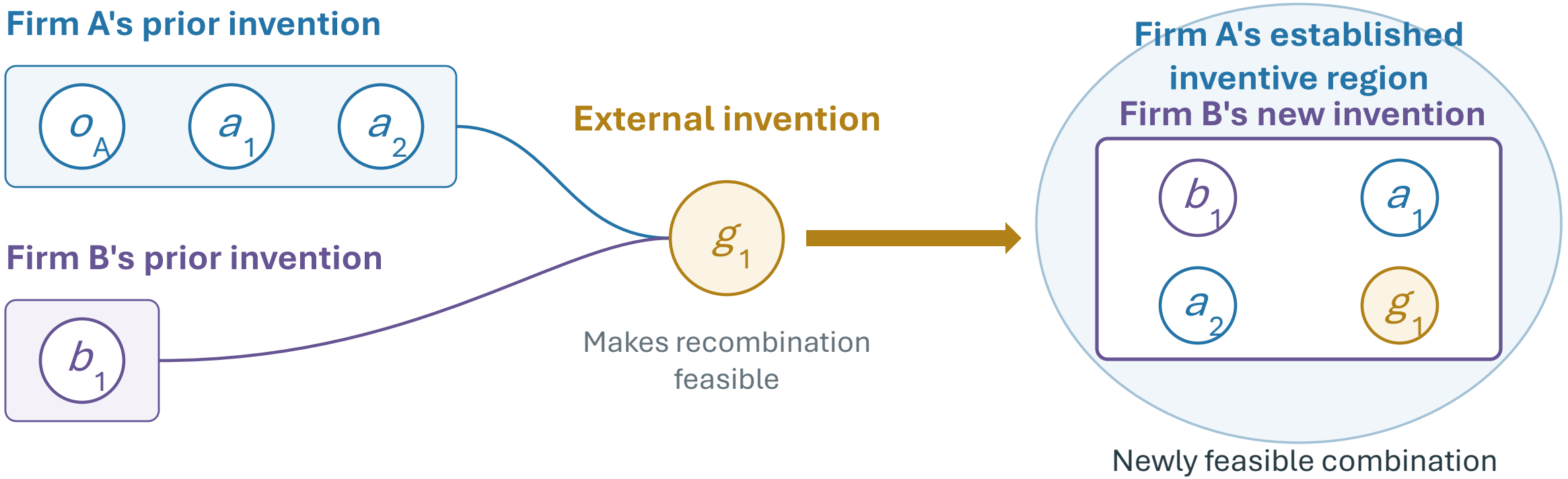
Dots: patents. Lines: citations.

- Search incursions are **competitive microinteractions** through which these transformations may emerge.

Inventions Enable Incursions

- Inventions can **enable new activities** or **lower the cost** of existing ones.

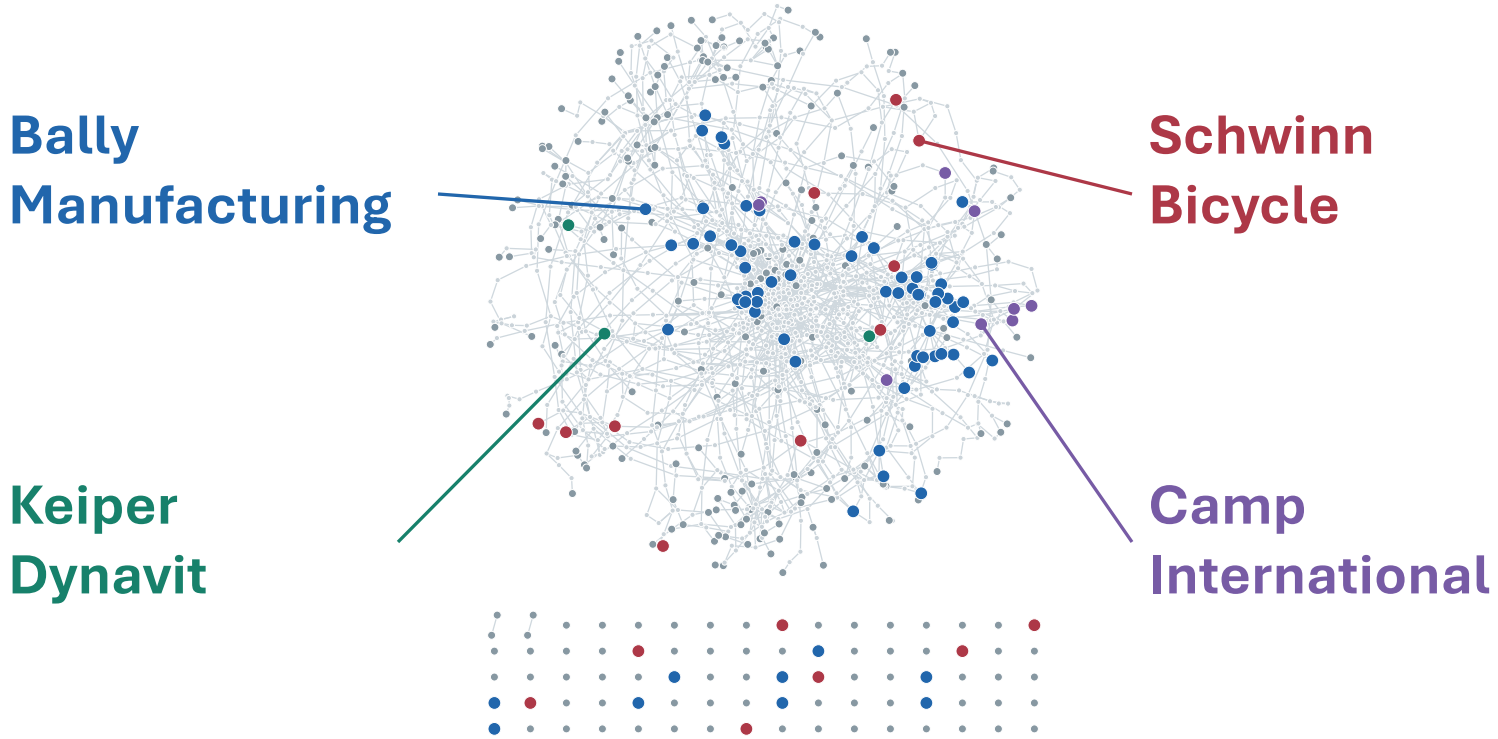
  (Schumpeter, 1934; Rosenberg, 1982)



Which Firms Gain an Advantage?

- An external invention may create **new possibilities for search incursions**.

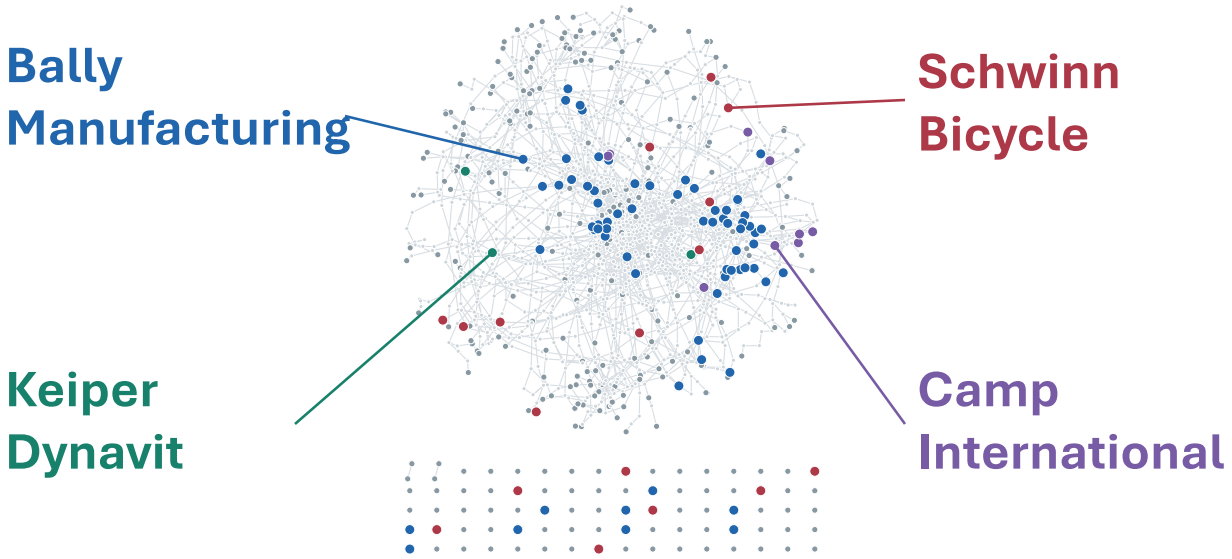
- **Which particular firms** gain opportunities to undertake incursions, and **which become vulnerable**?



Firms and Local Search

- **Firms search locally** for economic and behavioral reasons.
  (Cyert and March, 1963; Cohen and Levinthal, 1990; Stuart and Podolny, 1996; Ahuja and Lampert, 2001)
- Firms occupy **different positions** in technological knowledge networks.
  (Podolny, Stuart, and Hannan, 1996)
- These positions shape **interfirm search costs**.

Patent network



Directed search costs $\bar{C}$

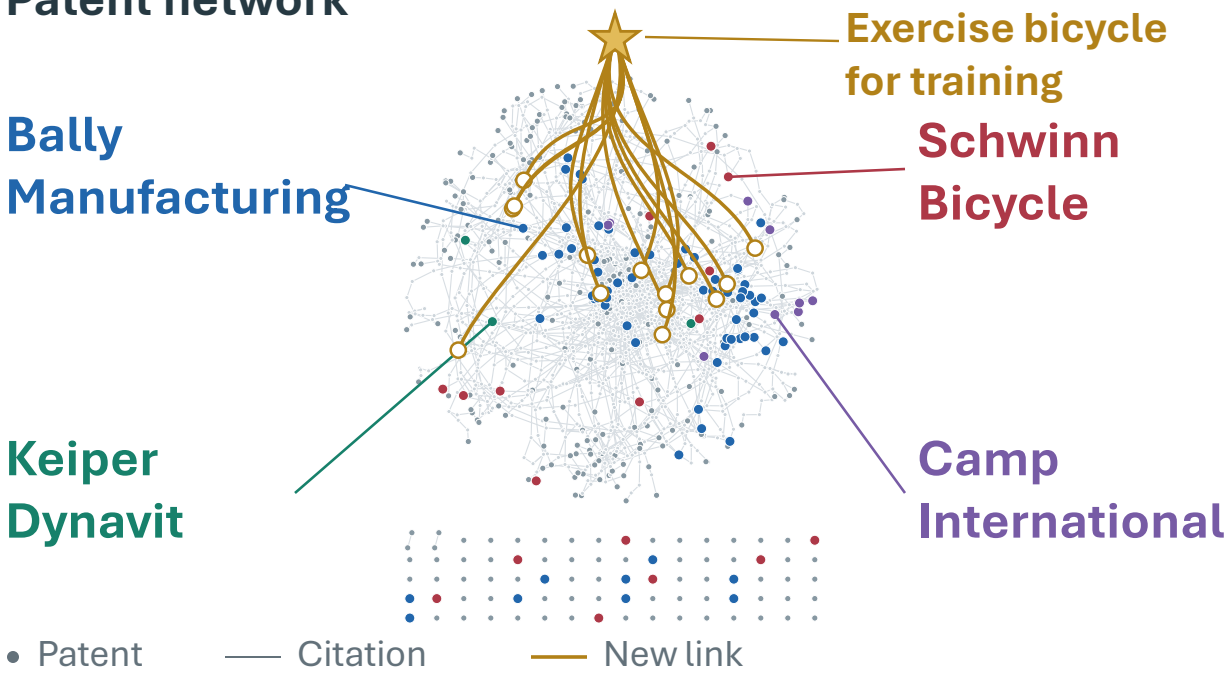
	Bally	Schwinn	Keiper	Camp
Bally		$\bar{C}_{12}$	$\bar{C}_{13}$	$\bar{C}_{14}$
Schwinn	$\bar{C}_{21}$		$\bar{C}_{23}$	$\bar{C}_{24}$
Keiper	$\bar{C}_{31}$	$\bar{C}_{32}$		$\bar{C}_{34}$
Camp	$\bar{C}_{41}$	$\bar{C}_{42}$	$\bar{C}_{43}$	

Rows: searching firm. Columns: destination.

Inventions Redefine Local Search

- Each invention **recombines existing knowledge elements**.
  (Fleming, 2001; Fleming and Sorenson, 2001)
- New connections reshape the knowledge network, changing **what is "local" for each firm**.

Patent network



Directed search costs C^g

	Bally	Schwinn	Keiper	Camp
Bally		C_{12}^g	C_{13}^g	C_{14}^g
Schwinn	C_{21}^g		C_{23}^g	C_{24}^g
Keiper	C_{31}^g	C_{32}^g		C_{34}^g
Camp	C_{41}^g	C_{42}^g	C_{43}^g	

Rows: searching firm. Columns: destination.

Relative Recombinant Advantage

C^{-}

	Bally	Schwinn	Keiper	Camp
Bally		c_{12}^{-}	c_{13}^{-}	c_{14}^{-}
Schwinn	c_{21}^{-}		c_{23}^{-}	c_{24}^{-}
Keiper	c_{31}^{-}	c_{32}^{-}		c_{34}^{-}
Camp	c_{41}^{-}	c_{42}^{-}	c_{43}^{-}	

C^{g}

	Bally	Schwinn	Keiper	Camp
Bally		c_{12}^{g}	c_{13}^{g}	c_{14}^{g}
Schwinn	c_{21}^{g}		c_{23}^{g}	c_{24}^{g}
Keiper	c_{31}^{g}	c_{32}^{g}		c_{34}^{g}
Camp	c_{41}^{g}	c_{42}^{g}	c_{43}^{g}	

Lower search costs mean greater accessibility.

Recombinant opportunity

How easily a firm can search toward other firms' inventive regions.

$$R_i^{g} / R_i^{-} = \frac{O_i^{g} / E_i^{g}}{O_i^{-} / E_i^{-}}$$

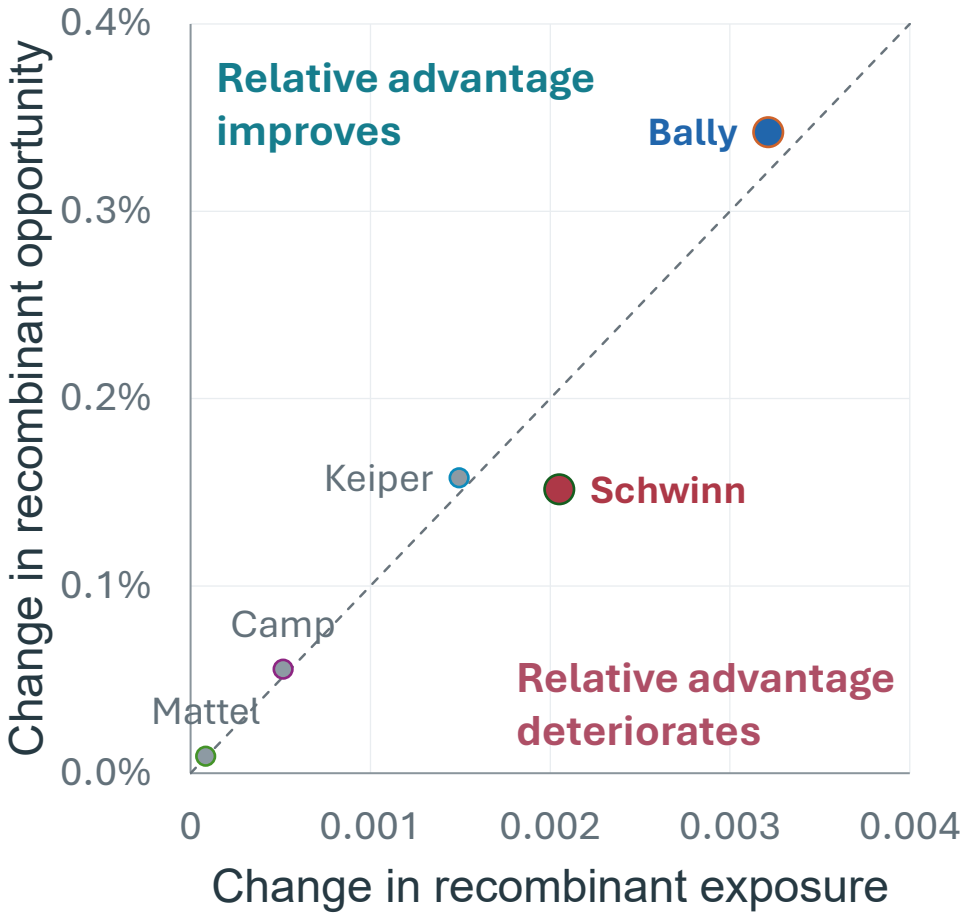
Recombinant exposure

How easily other firms can search toward its inventive region.

Change in the opportunity-to-exposure balance

Above 1: improves. **Below 1**: deteriorates.

One Invention, Different Recombinant Advantages



Bally Manufacturing

Opportunity **+0.342%**
Exposure **+0.321%**

Relative advantage improves

Schwinn Bicycle

Opportunity **+0.151%**
Exposure **+0.205%**

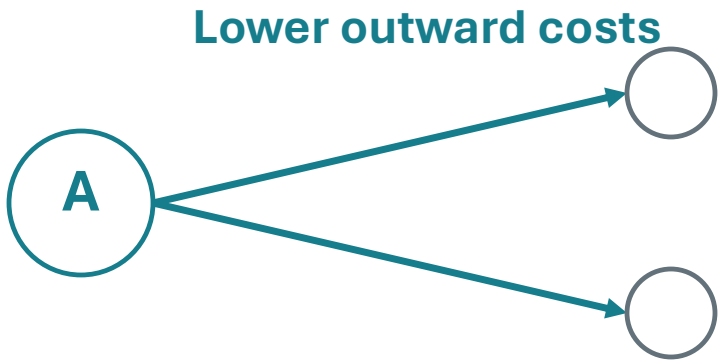
Relative advantage deteriorates

The same invention can change firms' **relative advantages in opposite directions**.

Firm-Level Relative Recombinant Advantage

Firm A: Undertaking incursions

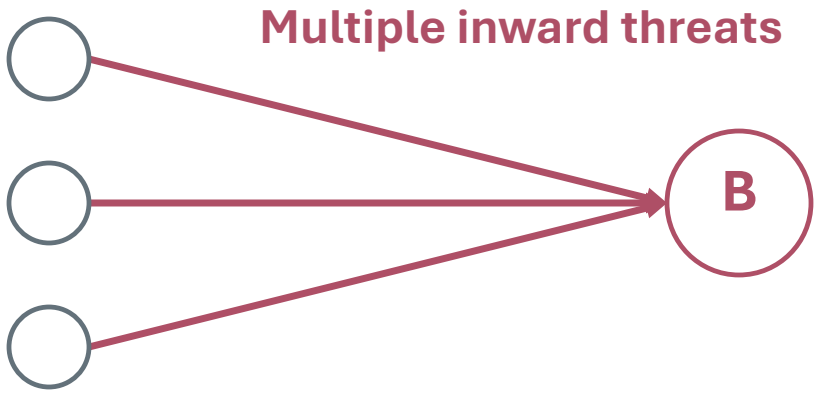
Unchanged defensive costs, combined with **new recombinant opportunities and lower outward search costs**, make incursions more attractive.



H1. Improvements in relative recombinant advantage increase the likelihood of **undertaking search incursions**.

Firm B: Becoming vulnerable

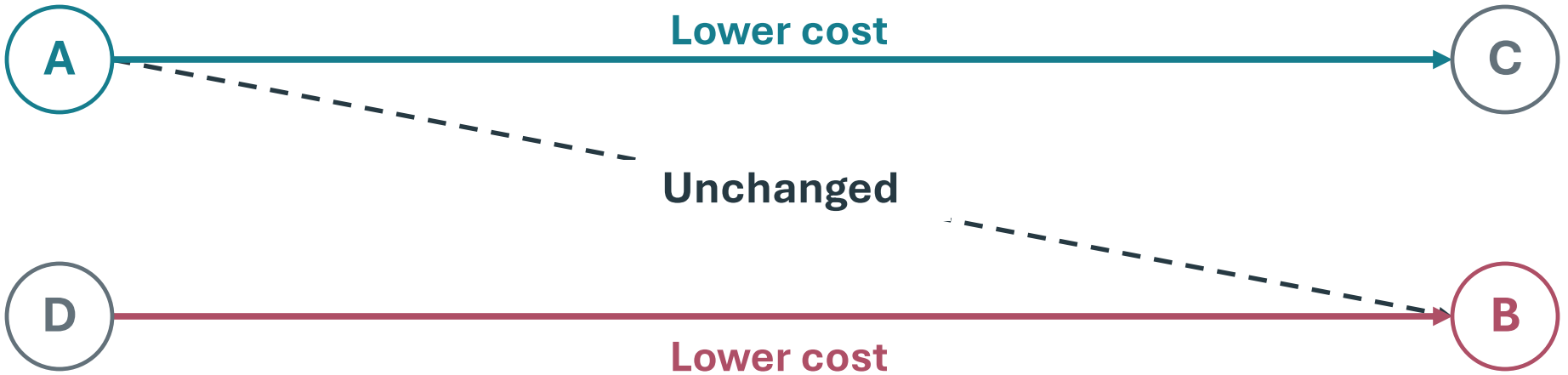
Firm B faces **multiple firm-specific, idiosyncratic threats**, placing competing demands on its inventive resources.



H2. Deteriorations in relative recombinant advantage increase the likelihood of **becoming an incursion target**.

From Firm-Level to Dyad-Level Advantage

- **Firm A's relative advantage improves**, perhaps through easier search toward Firm C.
- **Firm B's relative advantage deteriorates**, perhaps through greater exposure to Firm D.
- Yet **the cost of searching from Firm A toward Firm B may remain unchanged**.

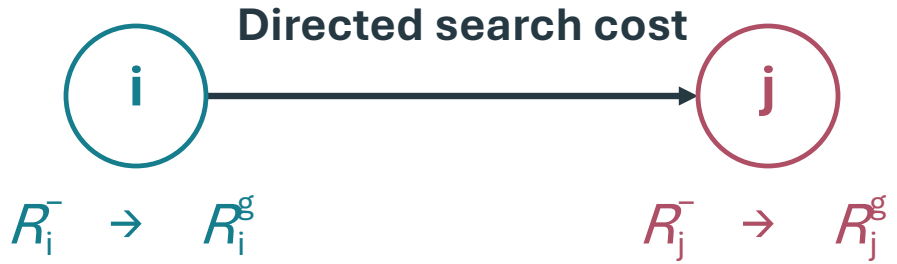


How has the invention changed the **directed search cost between these particular firms**?

Destination-Specific Recombinant Advantage

For each directed firm dyad, **Firm i to Firm j**, three invention-induced changes matter:

1. **Firm i's** opportunity-to-exposure balance.
2. **Firm j's** opportunity-to-exposure balance.
3. **The directed search cost** from Firm i toward Firm j.



$$\Delta C_{ij} = C_{ij}^g - C_{ij}^-$$

$$S_{ij}^s = \frac{1}{1 + C_{ij}^s} \frac{R_i^s}{R_j^s}$$

$$\Delta S_{ij} = \frac{S_{ij}^g}{S_{ij}^-} - 1$$

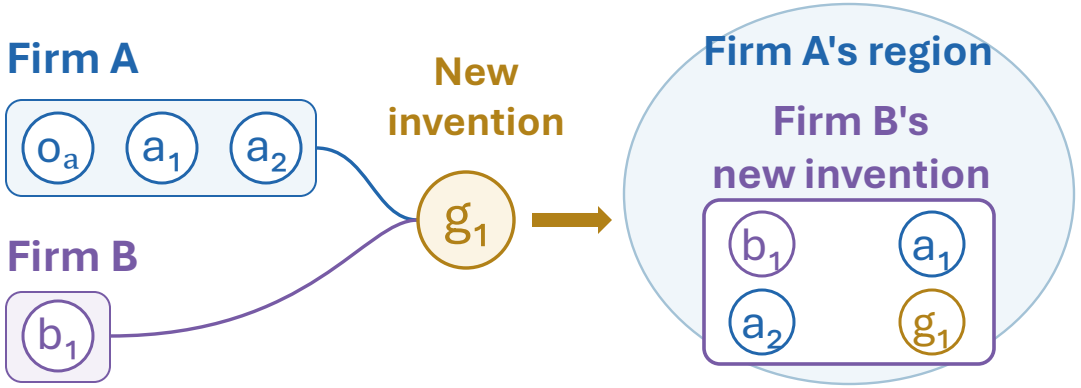
$R_i^s = O_i^s / E_i^s$ s = – (before), g (after)

H3. For a pair of firms *i* and *j*, the greater the invention-induced improvement in firm *i*'s **destination-specific recombinant advantage** toward firm *j*, the higher the probability that *i* enters *j*'s established inventive region, conditional on *i* undertaking a search incursion.

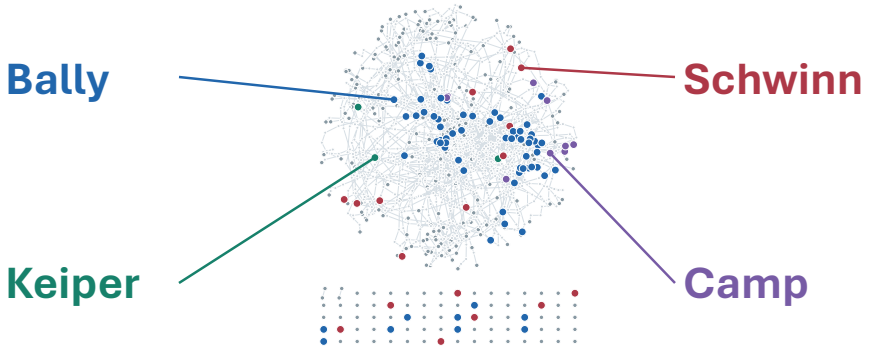
Unit of analysis: Directed firm dyads, conditional on the originating firm and focal invention.

Recap

1. Inventions enable search incursions



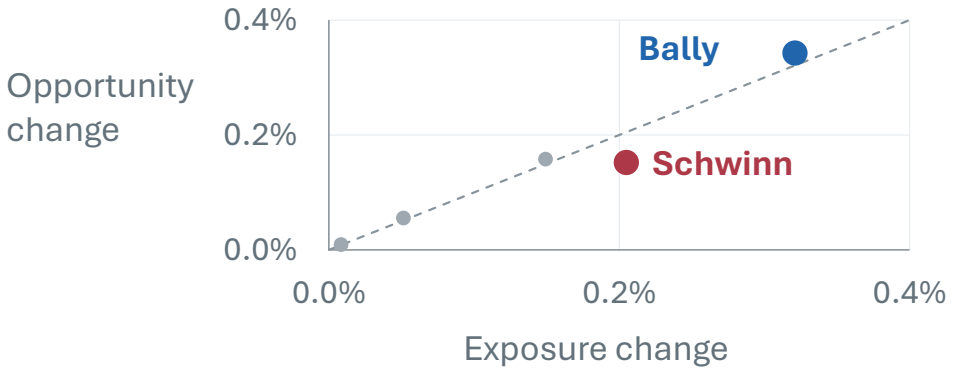
2. Technological positions determine search costs



3. Each invention changes these search costs



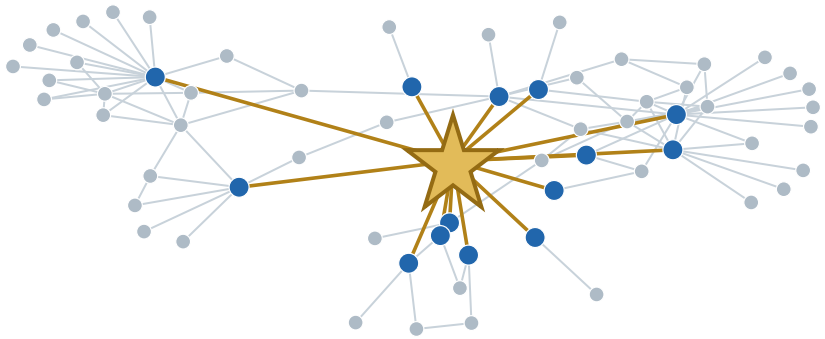
4. Some firms benefit, other firms become targets



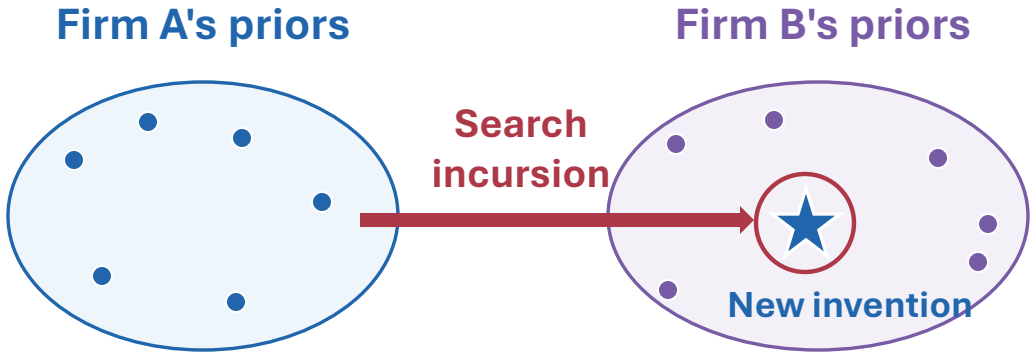
Data and Empirical Analysis

Empirical Overview

1. Identifying influential inventions



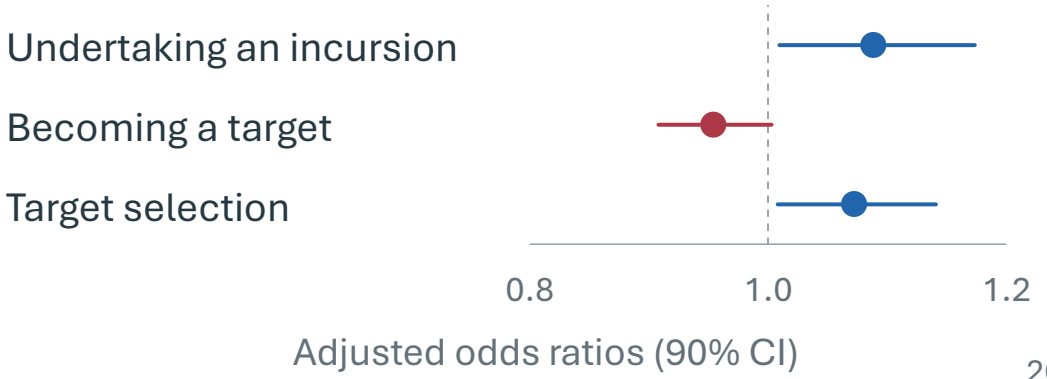
2. Identifying search incursions



3. Constructing technological networks before and after



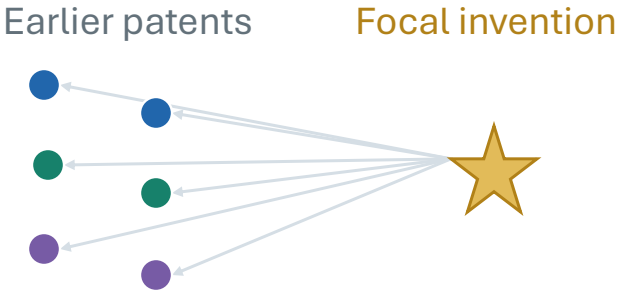
4. Inference



Identifying Influential Inventions

- Start with approximately **7.3 million U.S. utility patents**.
- Technique broadly based on **Rathje and Katila (2021)**.

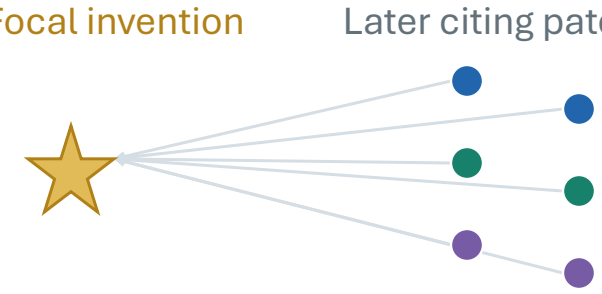
1. Upstream Technological Breadth



Diversity of backward-cited patents' three-digit USPC classes

$$1 - \sum_{c} s_{c}^{2}$$

2. Downstream Technological Breadth



15 years after grant

Diversity of forward-citing patents' three-digit USPC classes

$$1 - \sum_{c} s_{c}^{2}$$

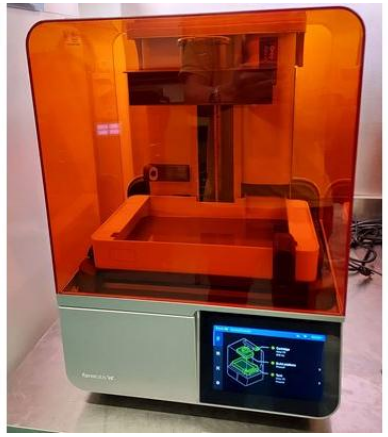
3. Distinct Downstream Use



Share of citing patents that do not also cite any predecessor

$$\frac{\text{Citers citing no predecessor}}{\text{All focal citers}}$$

Standardize the three components and average them to rank candidate inventions.



3D printing

Stereolithography, 1992



Touchscreens

Sliding-touch controls, 1991



Lithium-ion batteries

Carbon materials, 1999



Medical imaging

MRI vessel enhancement, 1997



Biotechnology

Gene-expression PCR, 1997



Robotics

Handheld motion control, 1997

Sample Overview

Five-year grant periods

	Patents	Share
1985–1989	786	15.3%
1990–1994	1,103	21.5%
1995–1999	1,271	24.8%
2000–2004	1,441	28.1%
2005–2009	522	10.2%

Technology sectors

	Patents	Share
Electrical engineering	2,343	45.7%
Multiple sectors	1,337	26.1%
Instruments	724	14.1%
Mechanical engineering	372	7.3%
Chemistry	288	5.6%
Other fields	54	1.1%
Unclassified	5	0.1%

5,123
Influential inventions

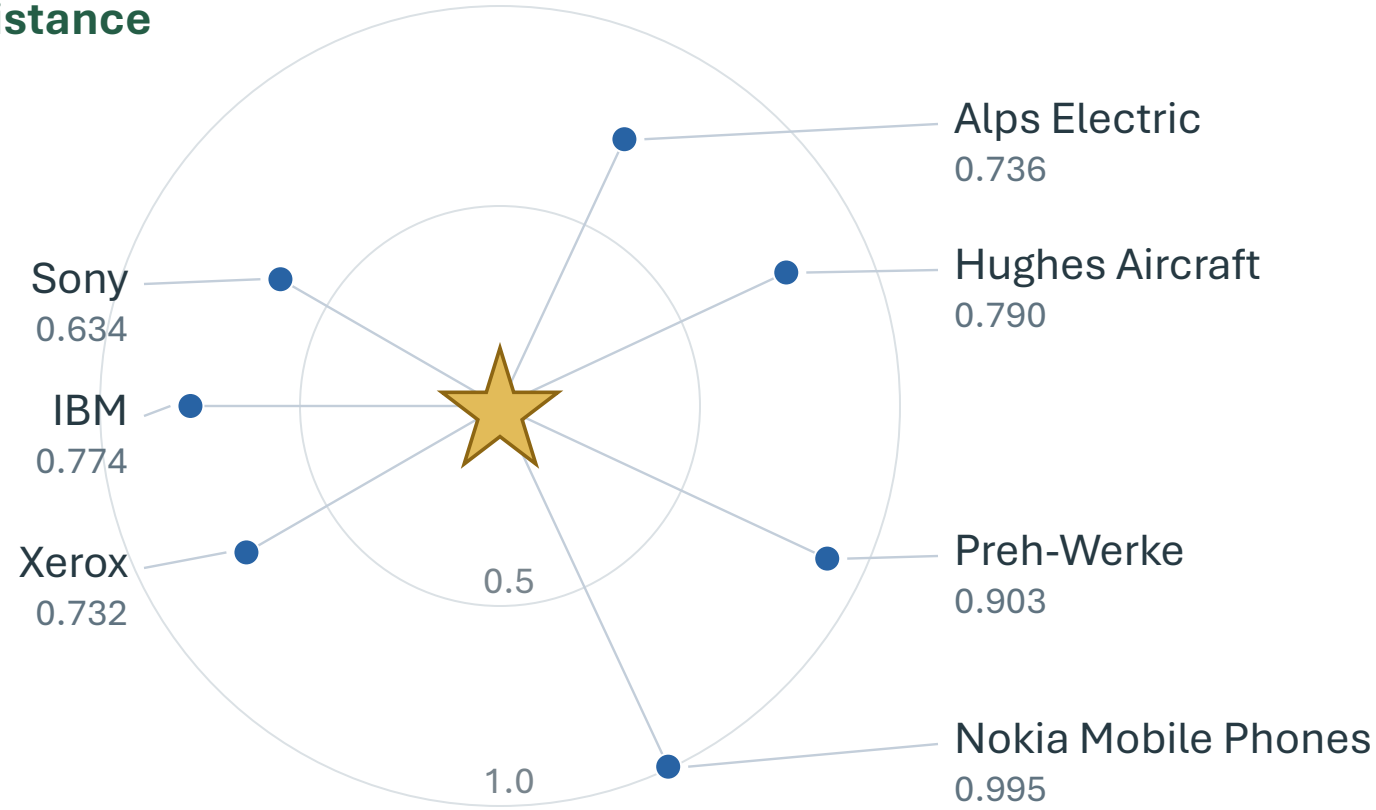
24,606
Distinct adopting firms

90,930
Invention–firm observations

Identifying Firms Building on the Invention

Touchscreen control panel with sliding touch control (1991)

Technological distance

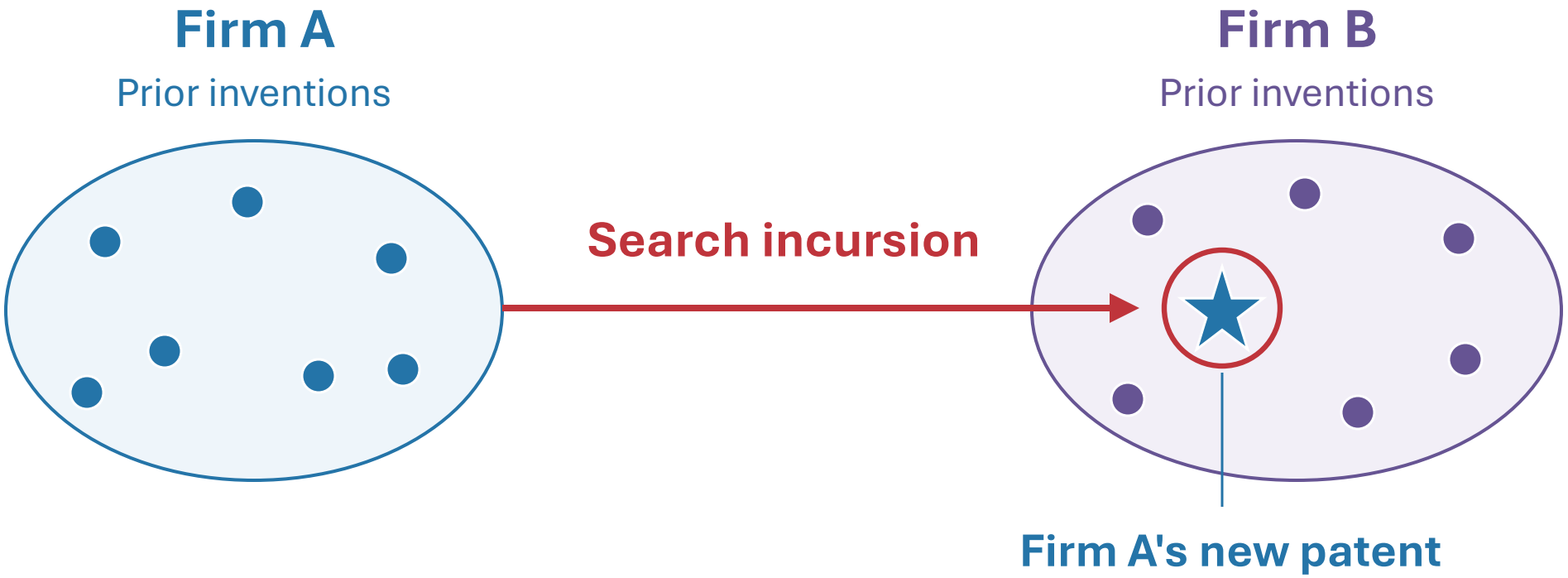


0 = same term profile 1 = no shared terms

We then gather all firms that built upon this invention within 15 years.

Identifying Search Incursions: I

- Identify firms that **subsequently cite** each influential invention.
- Reconstruct their inventive activity **before that invention**.
- Assess whether a citing patent **enters another firm's established inventive region**.



Identifying Search Incursions: II

1. Combine

Own knowledge



50:50

External knowledge

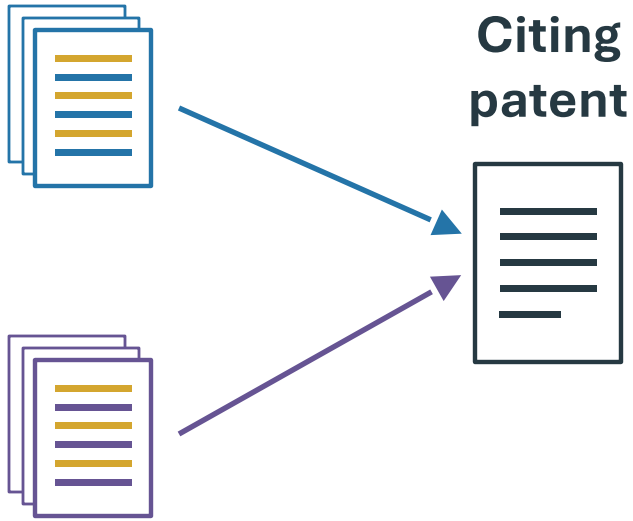


50:50

Up to 20 patents
from the preceding 15 years

2. Compare

1,000 comparisons

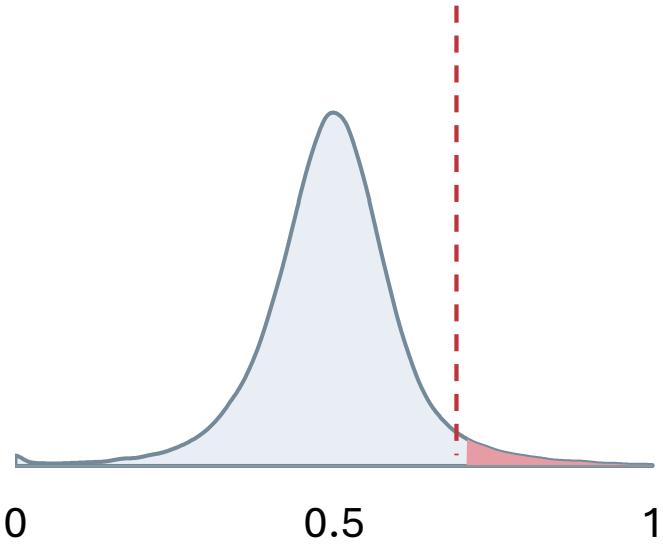


Cosine similarity

Score = share of external wins

3. Classify

0.709 cutoff

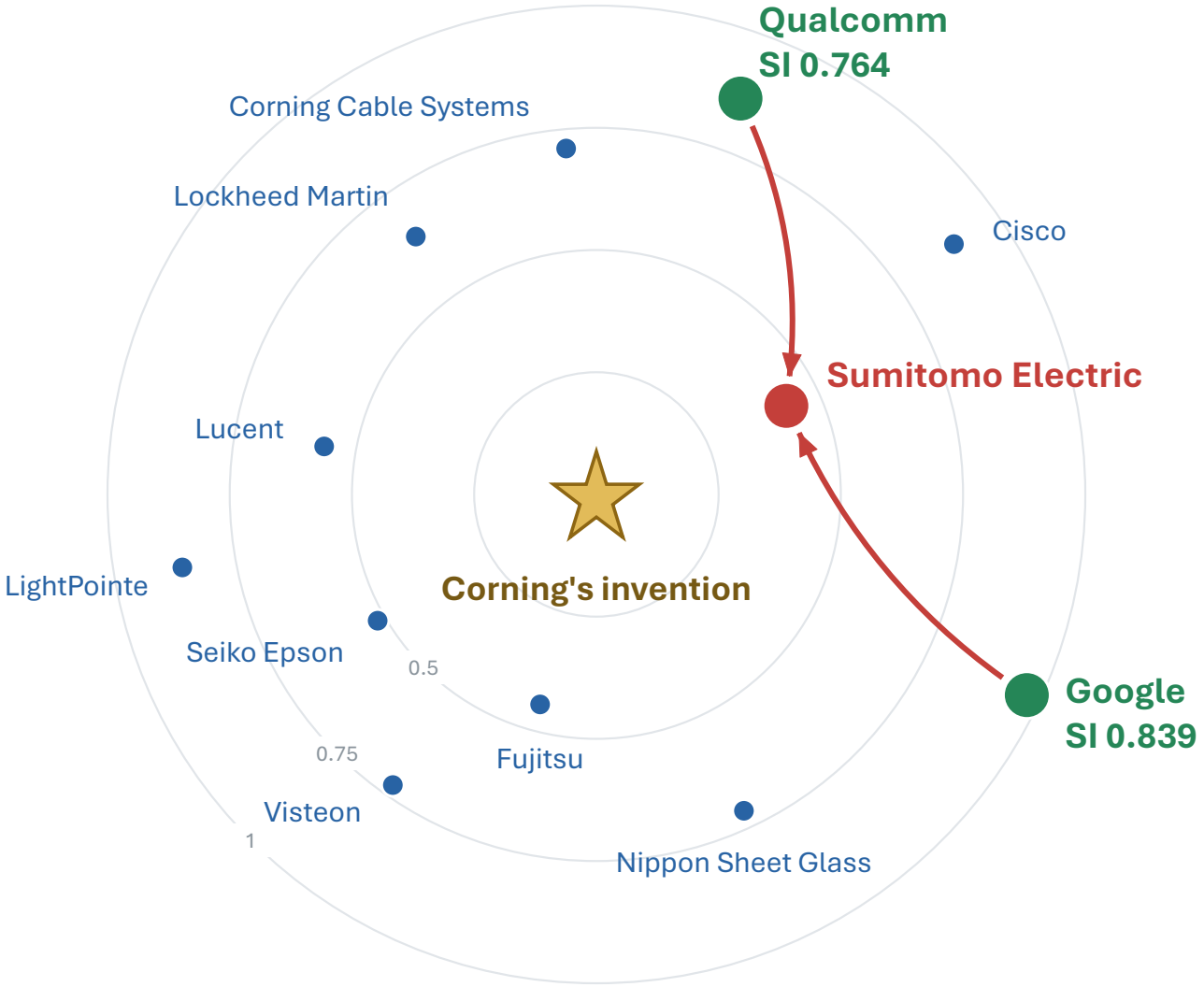


Score

Search incursions

Optical-Fiber Alignment

Corning | 2003 | US 6,587,618

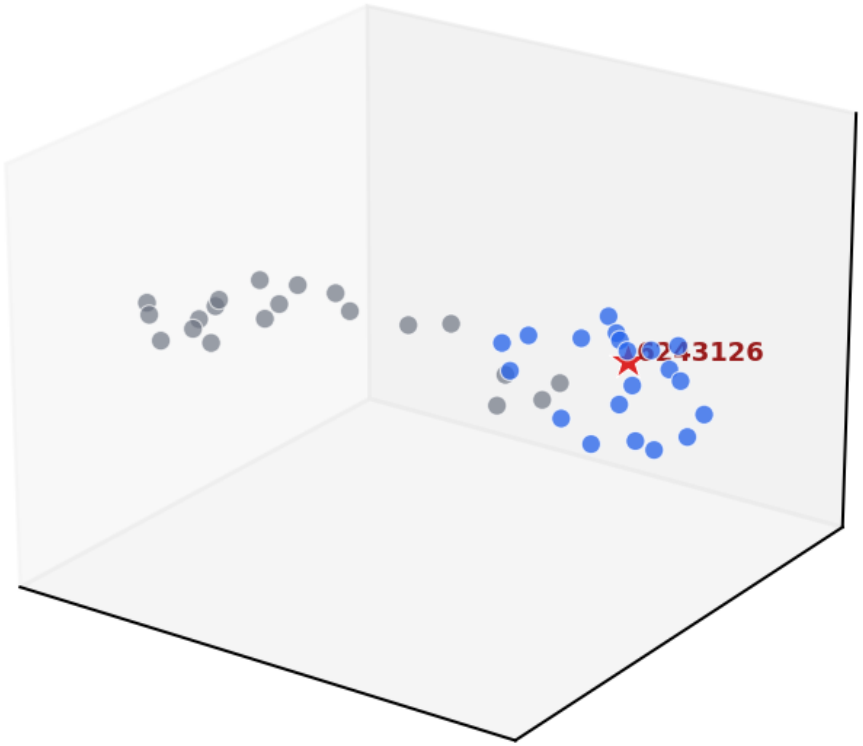


Radius: technological distance | 12 measured firms | SI cutoff: 0.709

Search Incursions

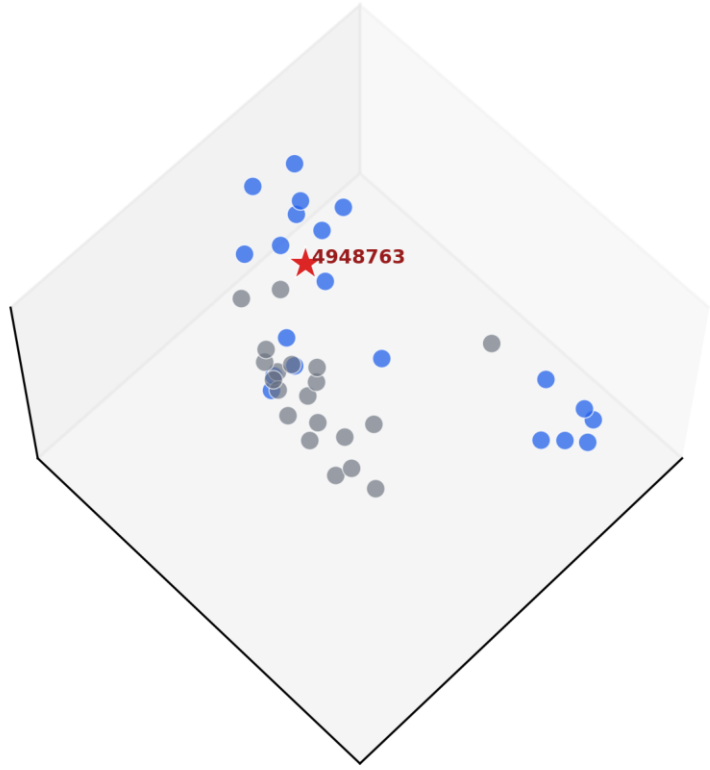
Laser image formation

US 6243126



Hollow ceramic fibers

US 4948763



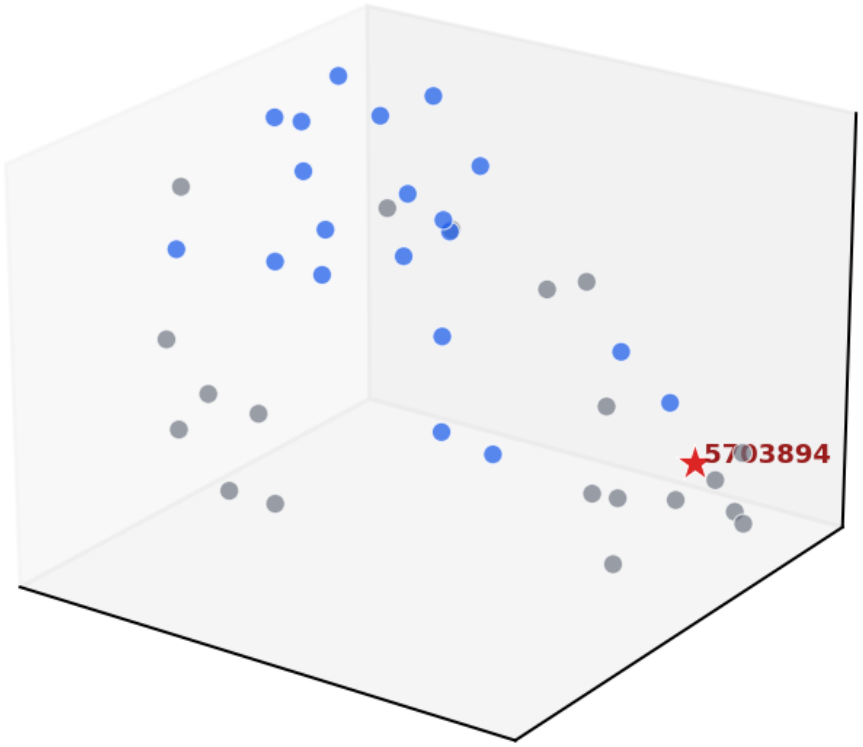
● Same-firm priors ● Other-firm priors ★ Focal patent

Earlier blinded-validation examples

Non-Incursions

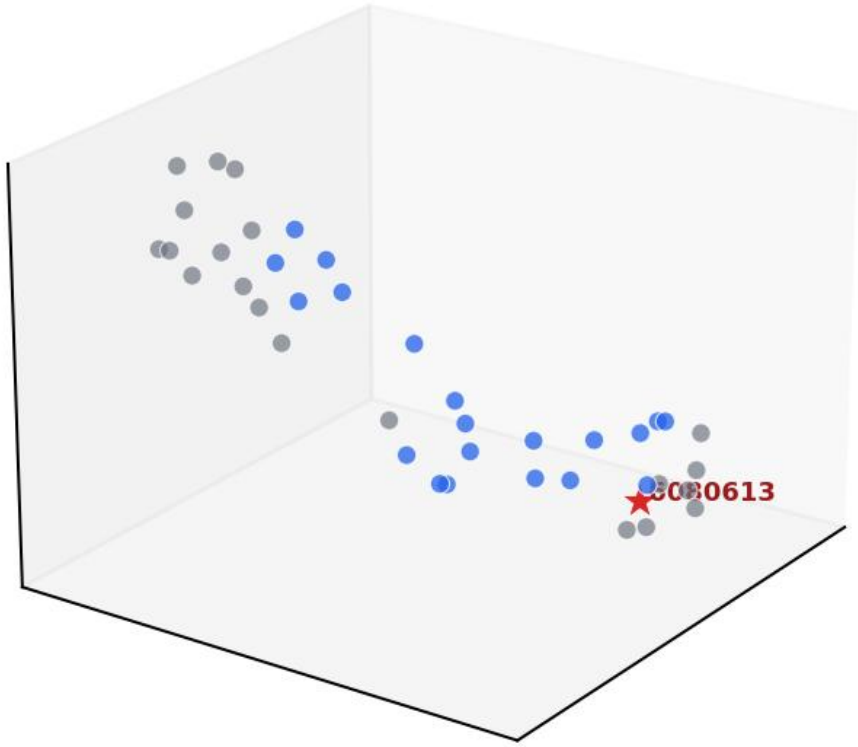
Semiconductor diodes

US 5703894



Integrated-circuit memory

US 6080613



● Same-firm priors ● Other-firm priors ★ Focal patent

Earlier blinded-validation examples

The Search-Incursion Sample

Incursions

Assessed invention–firm–patent links	139,644
Search-incursion events	**4,212 (3.02%)**
Distinct patents producing those events	3,888
Influential inventions with at least one incursion	1,727

Firms Involved

Distinct firms undertaking incursions	**1,346**
Distinct firms targeted	**1,197**
Distinct firms appearing in both roles	244

Corporate firms. Fifteen-year follow-up.
A patent may contribute multiple events. Firms may appear in both roles.

Firms' Prior Knowledge Positions

Comparing the two roles: Higher Lower

	All eligible firms	Undertaking incursions	Targeted firms
Prior patent stock (patents)	3,393	904	2,566
Patenting age (years)	13.4	6.9	12.2
Technological breadth (0–1)	0.775	0.353	0.724
Knowledge-base connectedness (%)	9.3	11.4	9.3
Textual distance to invention (0–1)	0.806	0.919	0.812
Recombinations new to firm (%)	47.6	75.9	52.1
Eligible firm–invention observations	51,113	2,186	2,573

Firms' Prior Search Behavior

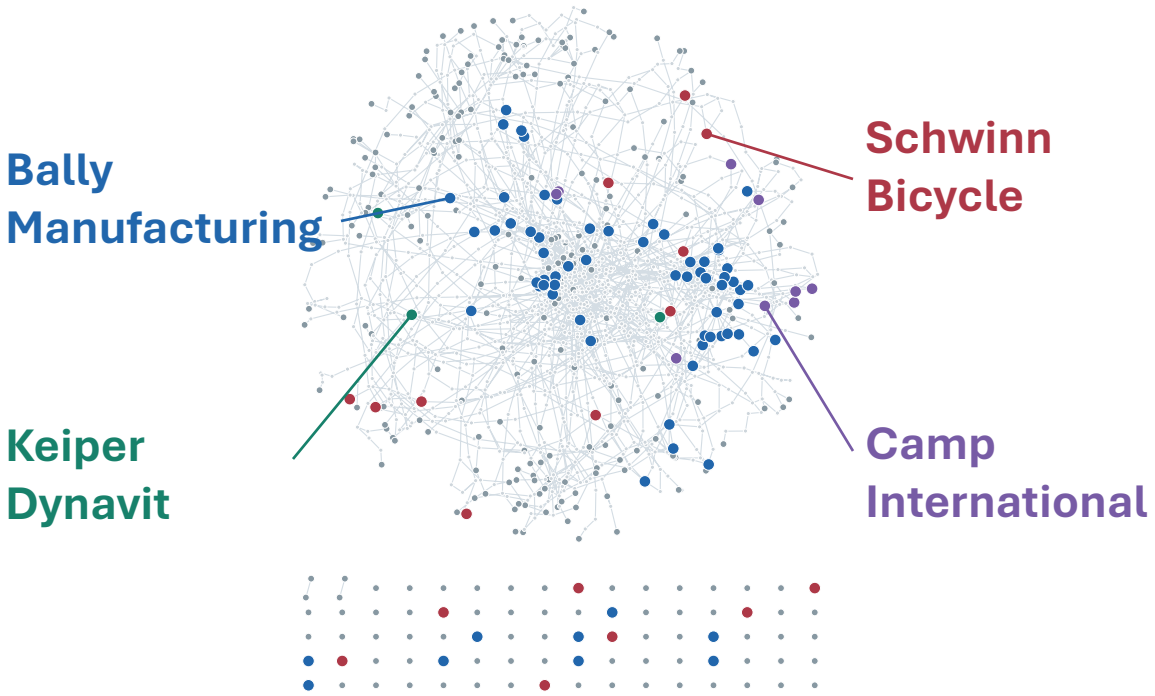
Comparing the two roles: Higher Lower

	All eligible firms	Undertaking incursions	Targeted firms
Search depth (prior uses per reference)	0.82	0.53	0.73
Search scope (%)	73.1	85.7	74.5
Prior invention originality (0–1)	0.405	0.425	0.401
External-citation share (%)	91.4	95.1	92.2
External-source nonredundancy (0–1)	0.502	0.465	0.520
Cross-firm, cross-class search (%)	51.8	55.6	51.9
Eligible firm–invention observations	51,113	2,186	2,573

Constructing Technological Knowledge Networks

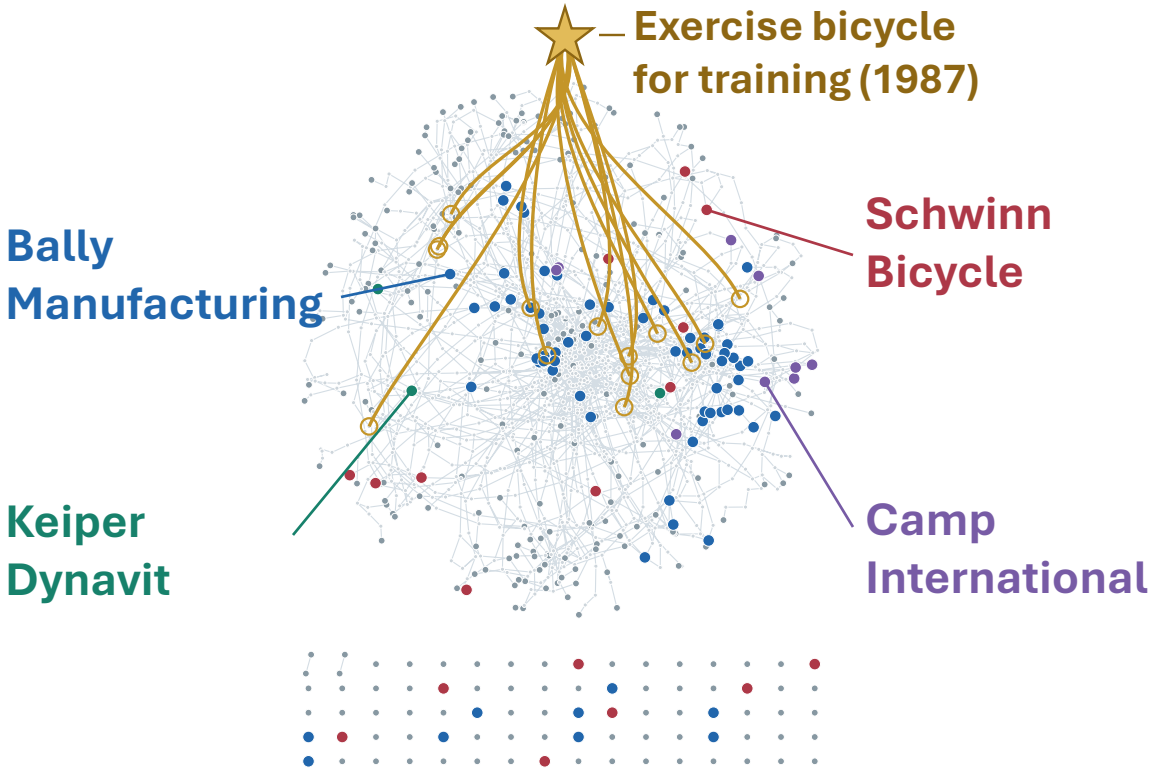
Before the invention

Patents from the preceding 15 years



After the invention

Add the invention and its citations



Nodes: patents. Links: citations. Colors: firms.

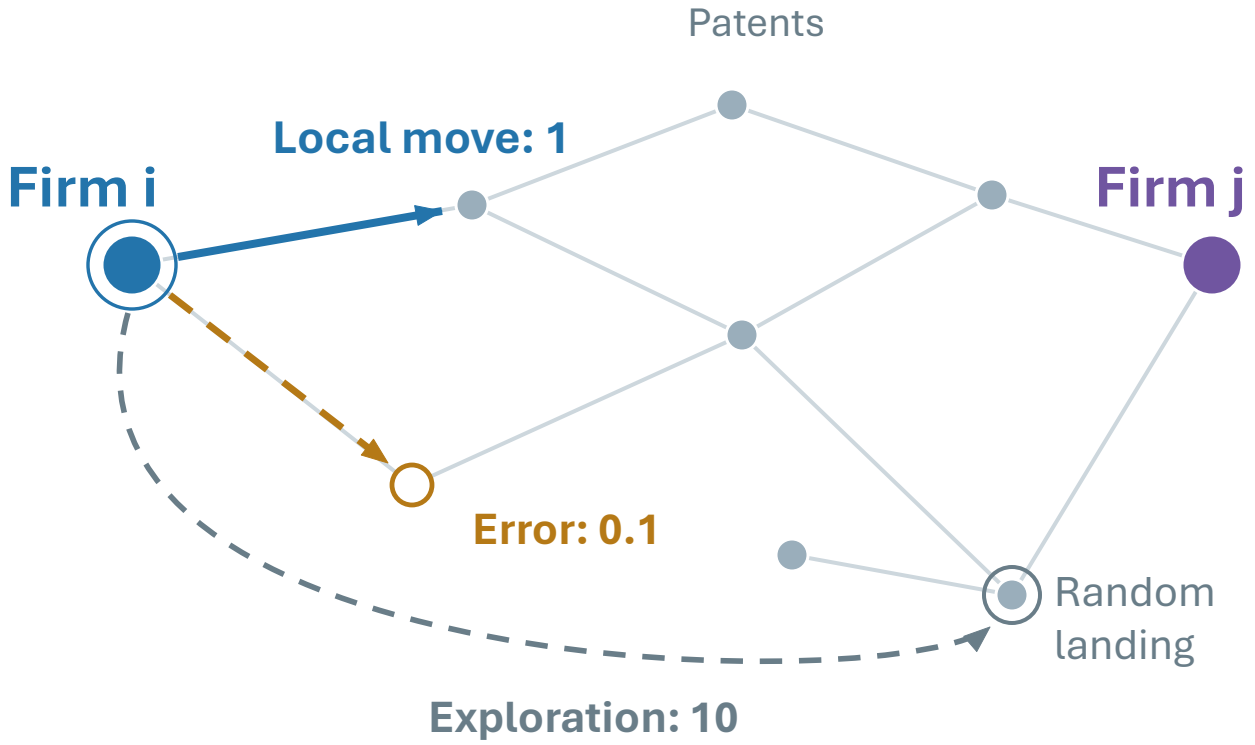
Measuring C_{ij}

How "local" is Firm j's knowledge base to Firm i?

- We conceptualize each firm as a **motivated walker**

  seeking the least costly way to reach another firm's knowledge.

- Each link traversed **1 unit**
- Navigation error **0.1**
- Exploratory jump **10 units**
- **Policy iteration (Howard, 1960)**



C_{ij} = average search cost across the two firms' prior patents

Lower C_{ij}: Firm j's knowledge is more local to Firm i.

Search-Cost Matrices

Exercise bicycle (1987): all prior patents

From the row firm to the column firm

Before: C^{-}

	Bally	Schwinn	Keiper	Camp	Mattel
Bally	-	2.866M	12.650	14.221	1.383M
Schwinn	0.838M	-	18.768	18.690	1.383M
Keiper	0.838M	2.866M	-	12.623	1.383M
Camp	0.838M	2.866M	12.664	-	1.383M
Mattel	0.838M	2.866M	15.576	16.201	-

After: C^{g}

	Bally	Schwinn	Keiper	Camp	Mattel
Bally	-	2.866M	**12.613**	**14.200**	1.383M
Schwinn	0.838M	-	18.768	**18.689**	1.383M
Keiper	0.838M	2.866M	-	12.623	1.383M
Camp	0.838M	2.866M	12.664	-	1.383M
Mattel	0.838M	2.866M	15.576	**16.200**	-

Bally's row: outward search costs

Bally's column: inward search costs

Bally to Keiper **12.650** to **12.613** average search cost

M = million cost units. Cells average all patent pairs; smaller changes may vanish after rounding.

Recombinant Opportunity and Exposure

Convert each patent-pair cost to accessibility, then average.

$$A_{ij}^{s} = \frac{1}{|P_i||P_j|} \sum_{p \in P_i} \sum_{q \in P_j} \frac{1}{1 + c_{pq}^{s}}$$

P: prior patents
s: before or after

$$O_i^s = \sum_{j \neq i} A_{ij}^s$$ **Sum the outward row**

$$E_i^s = \sum_{j \neq i} A_{ji}^s$$ **Sum the inward column**

$$R_i^s = O_i^s / E_i^s$$ **Opportunity / exposure**

Bally Manufacturing

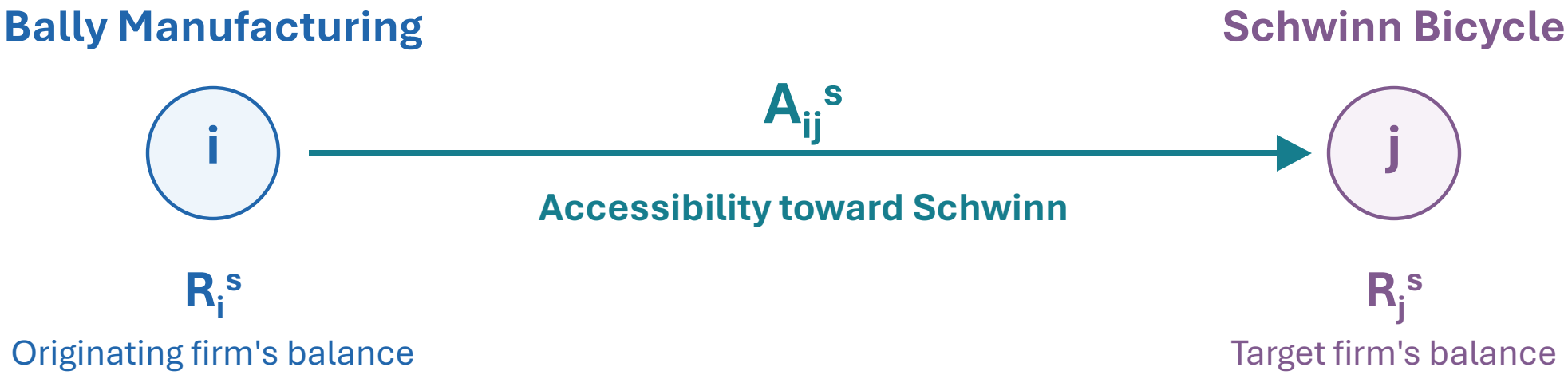
	Before	After
Opportunity	0.256258	0.257135
Exposure	0.262748	0.263592
Balance	0.975299	0.975503

$$\Delta R_i = \frac{R_i^g}{R_i^-} - 1 = \frac{0.97550289}{0.97529879} - 1 = +0.0209\%$$

Relative advantage improves

Destination-Specific Recombinant Advantage

A particular firm entering a particular other firm's inventive region



Destination-specific advantage

$$S_{ij}^{s} = A_{ij}^{s} \times \frac{R_i^s}{R_j^s}$$

Change after the invention

$$\Delta S_{ij} = \frac{S_{ij}^{g}}{S_{ij}^{-}} - 1$$

Higher ΔS_{ij}: greater improvement toward that particular firm.

Variables

	Dependent variable (0/1)	Independent variable
H1	Firm undertakes a search incursion	Change in relative recombinant advantage
H2	Firm becomes an incursion target	Change in relative recombinant advantage
H3	Firm i enters firm j's inventive region	Change in destination-specific recombinant advantage

Outcomes observed within 15 years of the influential invention.

Controls: Firm-specific, invention-specific, time, and dyad-specific factors.

Model: Logistic regression

Results: Undertaking an Incursion

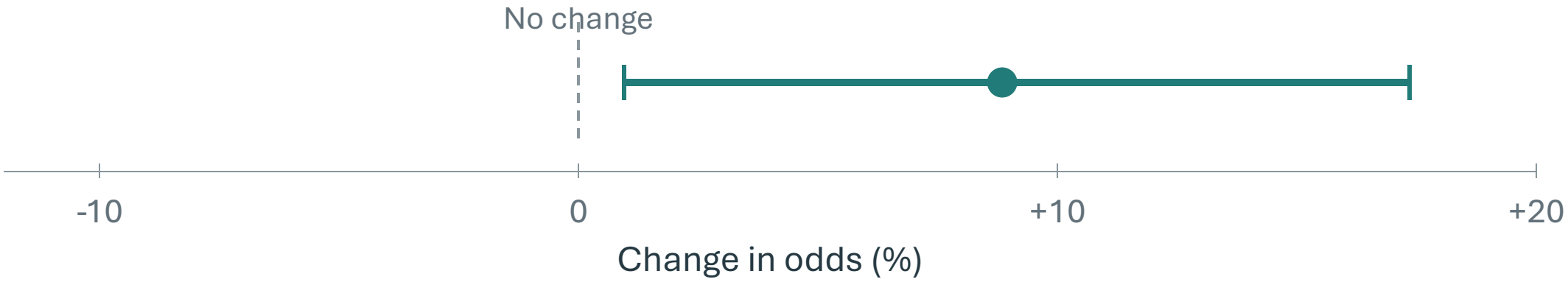
For a one-standard-deviation improvement in

relative recombinant advantage

8.8%

higher estimated odds of undertaking an incursion

90% interval: +1.0% to +17.4%



Adjusted logistic regression | N = 20,547

Invention fixed effects; uncertainty accounts for shared inventions and repeated firms.

Results: Becoming an Incursion Target

For a one-standard-deviation improvement in

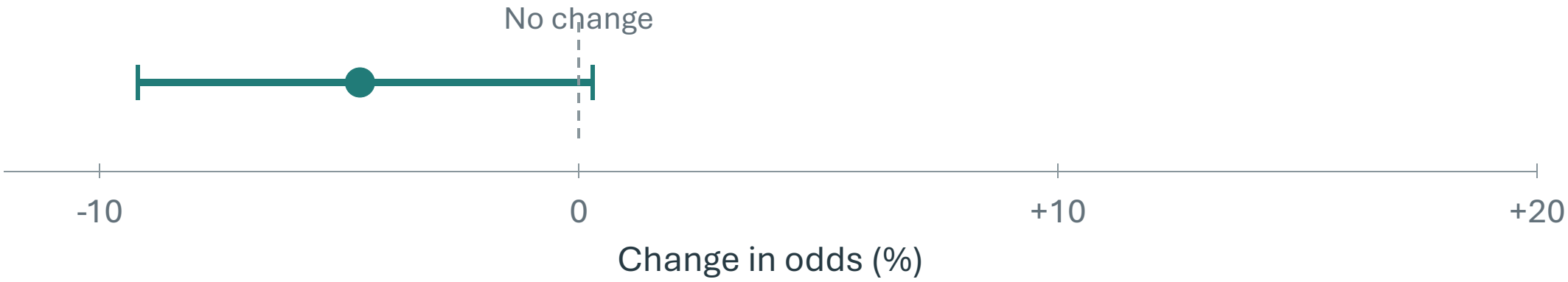
relative recombinant advantage

4.6%

lower estimated odds of becoming an incursion target

90% interval: -9.2% to +0.3%

Interval includes no change.



Adjusted logistic regression | N = 20,756

Invention fixed effects; uncertainty accounts for shared inventions and repeated firms.

Results: Selecting the Target

For a one-standard-deviation improvement in

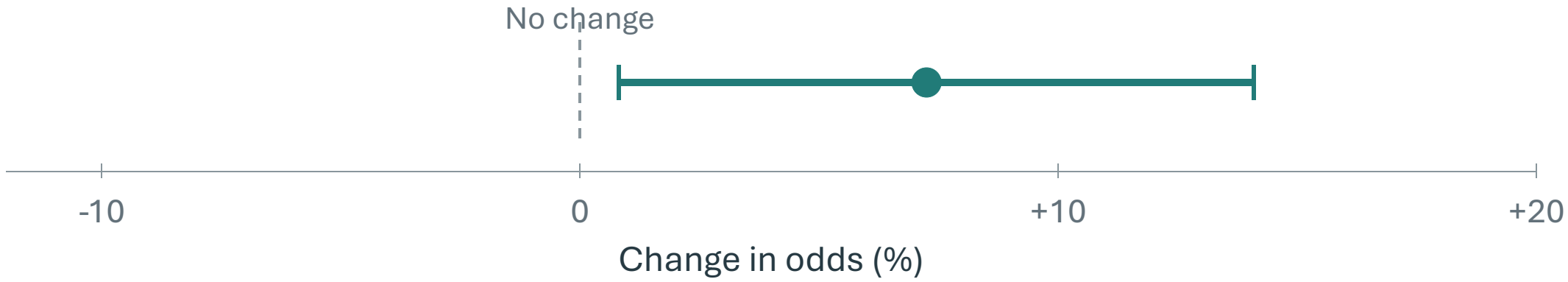
destination-specific recombinant advantage

7.2%

higher estimated odds of
entering that target's inventive region

90% interval: +0.8% to +14.1%

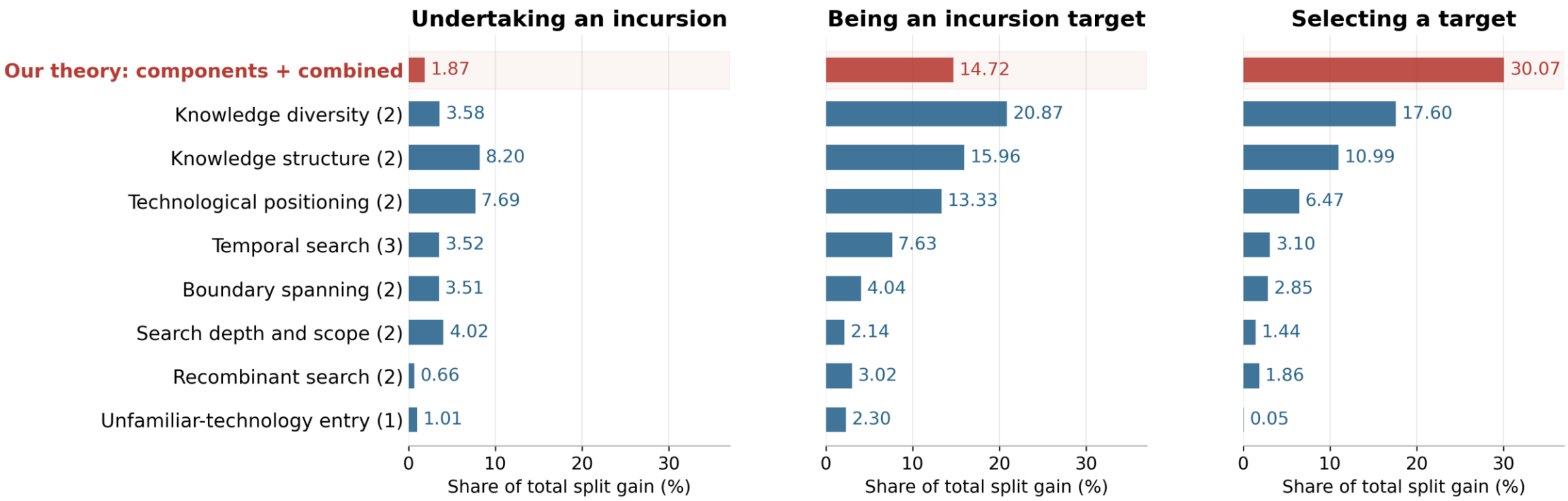
Among firms undertaking an incursion.



Adjusted logistic regression | N = 28,375

Origin-by-invention fixed effects; uncertainty accounts for shared inventions and firm endpoints.

Predictive Results



Our group: 3 columns in firm models; 8 in target selection. Other group sizes appear in parentheses.

XGBoost feature importance: share of total split gain.

Conclusions:

Toward a Relational Theory of Technological Competition

- Search incursions are **competitive microinteractions** through which broader technological transformations may emerge.
- An **invention-by-invention, firm-by-firm account** can reveal emerging inventive opportunities and **firm-specific, idiosyncratic threats** before their broader competitive consequences materialize.
- Our framework advances toward explaining **which particular firms gain inventive advantage over which particular other firms following each invention**.
- **Methodologically**, modeling inventive search as a **Markov decision process** and solving it through **policy iteration** offers a way to estimate how each invention changes otherwise unobserved, **directed interfirm search costs**.

Thank You

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Data and Case Sources

U.S. Patent and Trademark Office.
PatentsView. Patent records, citations, ownership, dates and technology classifications.

Hitachi, Ltd.
Corporate history.

Toyobo Co., Ltd.
Corporate history.

Google Patents.
Patent documents and bibliographic records for the illustrated inventions.

Justia Patents.
Patent documents for the motivating case and additional examples.

Appendix: Results and Supporting Evidence

Probability comparisons	**A1-A3**	Threshold and classification robustness	**A27-A33**
Full main regression estimates	**A4-A7**	Timing and influence	**A34-A39**
Model details	**A8**	Feature importance	**A40-A42**
Sample and descriptive results	**A9-A19**	Full threshold regressions	**A43-A50**
Adjusted relationships	**A20-A22**	Definitions	**A51-A52**
Mechanism components	**A23-A26**		

Select a section. All appendix slides are hidden during the main slideshow.

Adjusted Probabilities: Incursion Origin

Change in relative recombinant advantage

25th percentile

10.36%

90% CI: 9.98% to 10.76%

75th percentile

10.60%

90% CI: 10.18% to 11.03%

+0.239 percentage points

90% CI: +0.026 to +0.451 percentage points

Adjusted average probabilities; comparison group = invention.

25th versus 75th percentile within the same comparison group.

Adjusted Probabilities: Incursion Target

Change in relative recombinant advantage

25th percentile

12.05%

90% CI: 11.55% to 12.57%

75th percentile

11.84%

90% CI: 11.29% to 12.41%

-0.208 percentage points

90% CI: -0.426 to +0.009 percentage points

Adjusted average probabilities; comparison group = invention.

25th versus 75th percentile within the same comparison group.

Adjusted Probabilities: Target Selection

Change in destination-specific recombinant advantage

25th percentile

9.48%

90% CI: 9.09% to 9.88%

75th percentile

9.52%

90% CI: 9.12% to 9.93%

+0.040 percentage points

90% CI: +0.005 to +0.076 percentage points

Among firms undertaking an incursion; comparison group = origin x invention.

25th versus 75th percentile within the same comparison group.

Full Estimates

	H1 Incursion origin	H2 Incursion target	H3 Target selection
Relative recombinant advantage change	**0.085* (0.046)**	**-0.047 (0.030)**	**-**
Destination-specific advantage change	**-**	**-**	**0.070* (0.038)**
Odds ratio [90% CI]	1.088 [1.010, 1.174]	0.954 [0.908, 1.003]	1.072 [1.008, 1.141]
Observations	20,547	20,756	28,375
Positive outcomes	2,152	2,483	2,701
Fixed effects	Invention	Invention	Origin firm x invention
Uncertainty accounts for	Invention and firm	Invention and firm	Invention and shared firms

Logit coefficients (SE); odds ratios per 1 SD. * p < .10, ** p < .05, *** p < .01.
All control estimates follow. H3 is conditional on the origin undertaking an incursion.

Full Estimates: Firm Characteristics

	H1 Incursion origin	H2 Incursion target	H3 Target selection
Prior patent stock (log)	-0.196 (0.149)	-0.389*** (0.086)	-0.362*** (0.088)
Patenting age (log)	-0.129 (0.100)	0.097 (0.075)	0.071 (0.075)
Patenting momentum	0.032 (0.089)	-0.032 (0.062)	-0.062 (0.060)
Prior-patent originality	0.076 (0.046)	-0.043 (0.041)	-0.034 (0.040)
Knowledge-base connectedness	-0.060 (0.050)	0.005 (0.044)	0.026 (0.043)
One-patent portfolio	0.778*** (0.193)	-0.560*** (0.212)	-0.204 (0.207)
Technological breadth	0.087 (0.092)	-0.212*** (0.071)	-0.193*** (0.070)
Distance to the invention	0.217*** (0.080)	-0.206*** (0.038)	-0.222*** (0.036)

Coefficients (SE); continuous predictors per 1 SD. * p < .10, ** p < .05, *** p < .01.
H1/H2: focal firm; H3: potential target. One-patent portfolio: 0-to-1 comparison.

Full Estimates: Search and Network

	H1 Incursion origin	H2 Incursion target	H3 Target selection
Selected prior-patent count (log)	-1.421*** (0.145)	0.242*** (0.084)	0.115 (0.083)
Search depth (log)	0.326*** (0.106)	-0.126 (0.085)	-0.096 (0.090)
Search scope	-0.383*** (0.103)	-0.046 (0.079)	-0.042 (0.083)
External-citation share	-0.229*** (0.085)	-0.015 (0.042)	-0.0004 (0.038)
Knowledge-base vintage	0.118 (0.081)	-0.063 (0.054)	-0.068 (0.053)
External-source nonredundancy	-0.164*** (0.048)	0.081** (0.039)	0.071* (0.040)
Baseline network scale (log)	-1.249*** (0.212)	0.526*** (0.157)	0.535*** (0.181)
Baseline relative advantage (log)	-0.163*** (0.049)	-0.012 (0.058)	-0.185 (0.138)

Coefficients (SE); continuous predictors per 1 SD. * p < .10, ** p < .05, *** p < .01.
H1/H2: focal firm; H3: potential target. One-patent portfolio: 0-to-1 comparison.

Full Estimates: Dyads and Missing Data

	H1 Incursion origin	H2 Incursion target	H3 Target selection
Prior citations: origin to target	-	-	-0.158 (0.122)
Prior citations: target to origin	-	-	-0.140 (0.092)
Baseline directed accessibility (log)	-	-	-0.179 (0.153)
Missing originality	0.259 (0.665)	-0.654 (0.849)	-0.581 (0.843)
Missing technological breadth	0.161 (0.336)	0.534** (0.263)	0.363 (0.260)
Missing search depth	Omitted	Omitted	Omitted
Missing search scope	-0.283 (0.935)	1.047 (1.122)	0.697 (1.301)
Missing external-citation share	0.152 (0.519)	0.869 (0.607)	1.009 (0.616)
Missing source nonredundancy	-0.302 (0.307)	-0.569 (0.372)	-0.367 (0.334)

Logit coefficients (SE). * p < .10, ** p < .05, *** p < .01.
Omitted: collinear after fixed effects. A dash means the variable is not included.

Models and Statistical Inference

H1 and H2

Compare firms around the same invention.

Invention fixed effects; dependence by invention and firm.

H3

Compare potential targets for the same origin and invention.

Origin-by-invention fixed effects; shared-invention and shared-firm dependence.

Interpretation

Adjusted associations within the selected incursion-invention cohort.

Continuous estimates use one fixed pooled-design standard deviation.

Profiled fixed-intercept logistic regression. 90% confidence intervals.

Influential Inventions: Coverage

A. Coverage

Characteristic	Corporate adoption universe	Full network cohort
Influential inventions	5,123	1,727
Distinct adopting firms	24,606	4,824
Firm-invention observations	90,930	21,646
invention score	0.63 (0.19)	0.61 (0.19)
Adopting firms per invention	17.75 (12.87)	12.53 (8.12)
Inventors per invention	2.55 (1.78)	2.55 (1.77)
Backward technological breadth	0.74 (0.12)	0.74 (0.12)

Continuous entries are mean (SD). Technology sectors describe patents.

Influential Inventions: Grant Cohorts

B. Grant cohort: number of inventions (%)

Characteristic	Corporate adoption universe	Full network cohort
1985-1989	786 (15.3%)	229 (13.3%)
1990-1994	1,103 (21.5%)	397 (23.0%)
1995-1999	1,271 (24.8%)	389 (22.5%)
2000-2004	1,441 (28.1%)	523 (30.3%)
2005-2009	522 (10.2%)	189 (10.9%)

Continuous entries are mean (SD). Technology sectors describe patents.

Influential Inventions: Technologies

C. Technology sector: number of inventions (%)

Characteristic	Corporate adoption universe	Full network cohort
Electrical engineering	2,343 (45.7%)	878 (50.8%)
Multiple sectors	1,337 (26.1%)	430 (24.9%)
Instruments	724 (14.1%)	229 (13.3%)
Mechanical engineering	372 (7.3%)	99 (5.7%)
Chemistry	288 (5.6%)	72 (4.2%)
Other fields	54 (1.1%)	18 (1.0%)
Unclassified	5 (0.1%)	1 (0.1%)

Continuous entries are mean (SD). Technology sectors describe patents.

Adoption Sample

A. Corporate adoption universe

Unit or statistic	Count
inventions	5,123
Distinct adopting firms	24,606
Firm-invention observations	90,930
Adoption links (invention-firm-patent)	263,680
Scored adoption links	139,644
Firm-invention observations with at least one score	50,658

Patents, adoption links, firms and directed pairs are distinct counting units.

Observed Search Incursions

B. Elbow-defined incursion activity

Unit or statistic	Count
inventions with an observed incursion	1,727
Distinct SI patents	3,888
SI adoption links	4,212
Distinct origin firms	1,346
Distinct attributed target firms	1,197
Distinct firms observed in both roles	244
Distinct invention-origin-target pairs	2,773

Patents, adoption links, firms and directed pairs are distinct counting units.

Network and Regression Coverage

C. Network and regression coverage

Unit or statistic	Count
Full-network firm-invention observations	21,646
Full-network directed pairs	363,390
Retained firm-invention descriptive observations	21,102
Retained observations with known origin status	20,725
Retained known-origin directed pairs	349,353

Patents, adoption links, firms and directed pairs are distinct counting units.

Network Positions Before the Invention

A. Firm-invention positions before insertion

Measure	N	Mean	SD	Median
Opportunity: outward accessibility	21,102	2.734	2.388	2.168
Exposure: inward accessibility	21,102	2.734	2.393	2.169
Relative advantage: opportunity / exposure	21,102	2,210.766	37,791.730	1.000

bp = basis points of proportional change. 1 bp = 0.01%.

Invention-Induced Network Changes

B. Invention-induced changes

Measure	N	Mean	SD	Median
Opportunity change (%)	21,102	1.065	2.637	0.011
Exposure change (%)	21,102	1.061	2.643	0.003
Firm relative-advantage change (bp)	21,102	0.467	7.438	0.022
Directed accessibility change (bp)	349,353	109.387	483.137	0.000
Directed-advantage change (bp)	349,353	109.964	483.132	0.025

bp = basis points of proportional change. 1 bp = 0.01%.

Firm Roles: Network and Firm Characteristics

Characteristic	All firms	Origin only	Target only	Both	Neither
Firm-invention observations	21,102	2,123	2,402	36	16,164
Relative-advantage change (bp)	0.47 (7.44)	0.73 (5.37)	0.09 (7.98)	1.38 (5.16)	0.50 (7.57)
Opportunity change (%)	1.07 (2.64)	0.41 (1.66)	1.07 (2.76)	0.17 (0.47)	1.15 (2.71)
Exposure change (%)	1.06 (2.64)	0.40 (1.65)	1.07 (2.77)	0.16 (0.45)	1.14 (2.71)
Prior patent stock	3,063 (6,378)	853 (3,881)	2,519 (5,791)	1,184 (5,582)	3,455 (6,675)
Patenting age (years)	12.7 (8.7)	6.8 (7.5)	12.2 (8.2)	5.3 (7.6)	13.6 (8.5)

Mean (SD). Roles are mutually exclusive within each invention.

Firm Roles: Knowledge and Search

Characteristic	All firms	Origin only	Target only	Both	Neither
Patenting momentum	0.90 (1.14)	0.38 (0.65)	0.89 (1.10)	0.37 (0.59)	0.98 (1.18)
Prior-patent originality	0.41 (0.16)	0.42 (0.23)	0.40 (0.16)	0.42 (0.25)	0.41 (0.15)
Knowledge-base connectedness	0.10 (0.17)	0.11 (0.22)	0.09 (0.17)	0.05 (0.13)	0.09 (0.17)
Technological breadth	0.74 (0.30)	0.35 (0.37)	0.73 (0.28)	0.20 (0.34)	0.79 (0.25)
Semantic distance to invention	0.82 (0.13)	0.92 (0.11)	0.81 (0.14)	0.92 (0.09)	0.81 (0.13)
Prior search scope	0.74 (0.22)	0.86 (0.23)	0.74 (0.21)	0.89 (0.21)	0.73 (0.21)

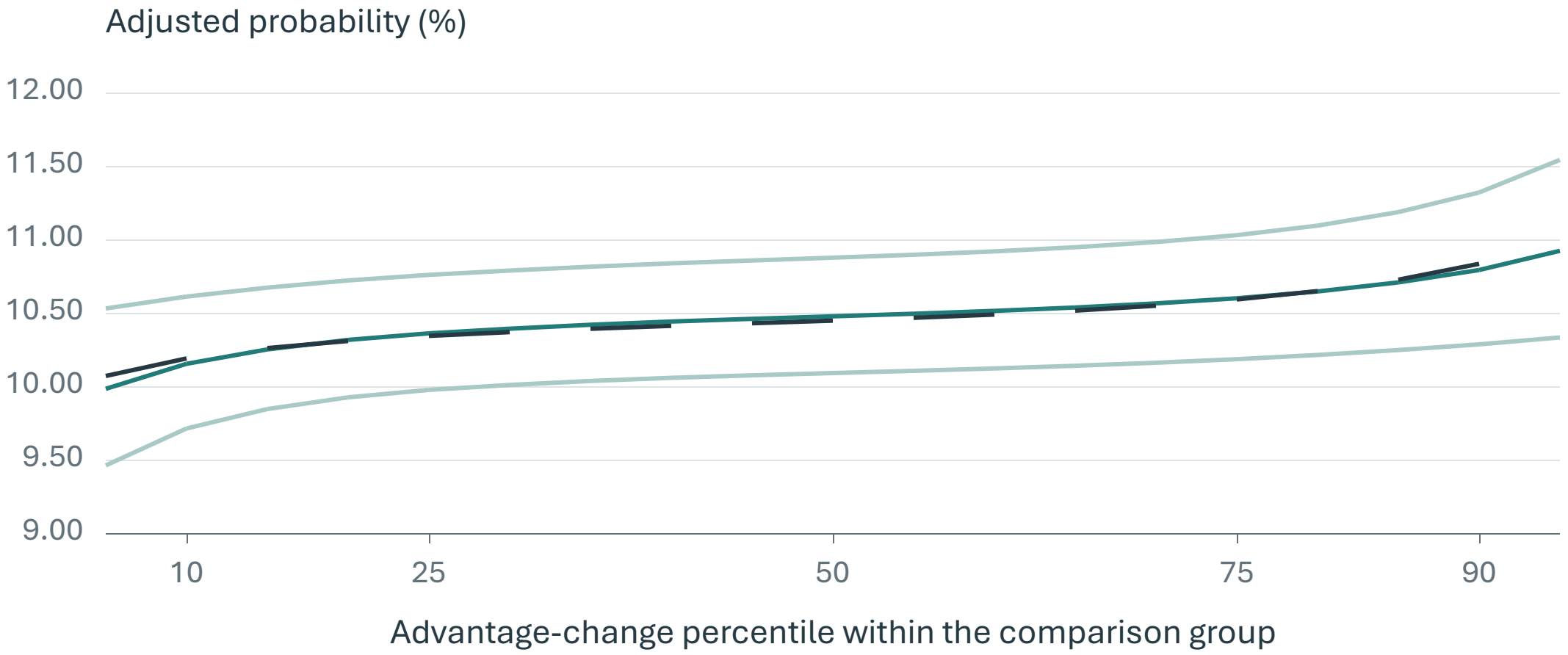
Mean (SD). Roles are mutually exclusive within each invention.

Firm Roles: Outcome Coverage

Characteristic	All firms	Origin only	Target only	Both	Neither
Unavailable origin status	377	0	0	0	0

Mean (SD). Roles are mutually exclusive within each invention.

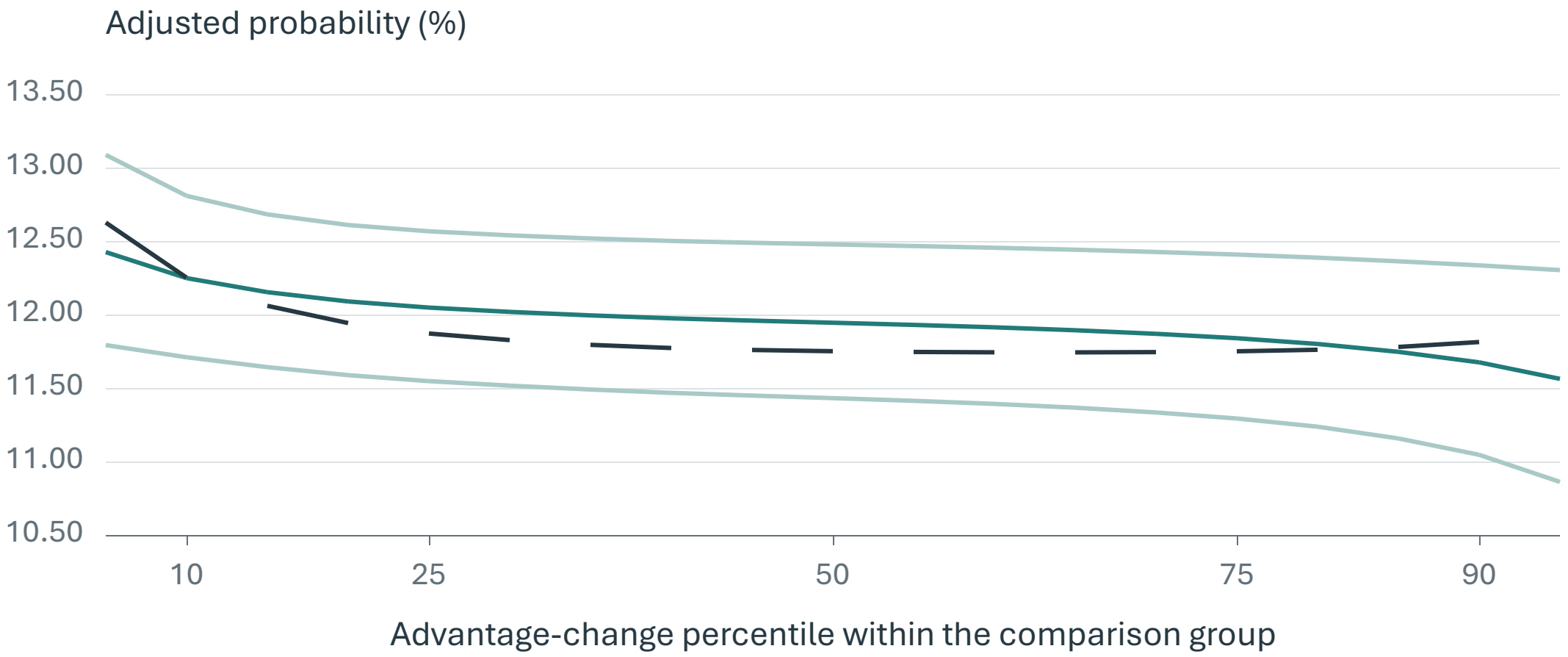
Adjusted Relationship: Incursion Origin



Nonlinearity test: p = 0.693

Solid: main logit; dashed: flexible specification. Pointwise 90% intervals.

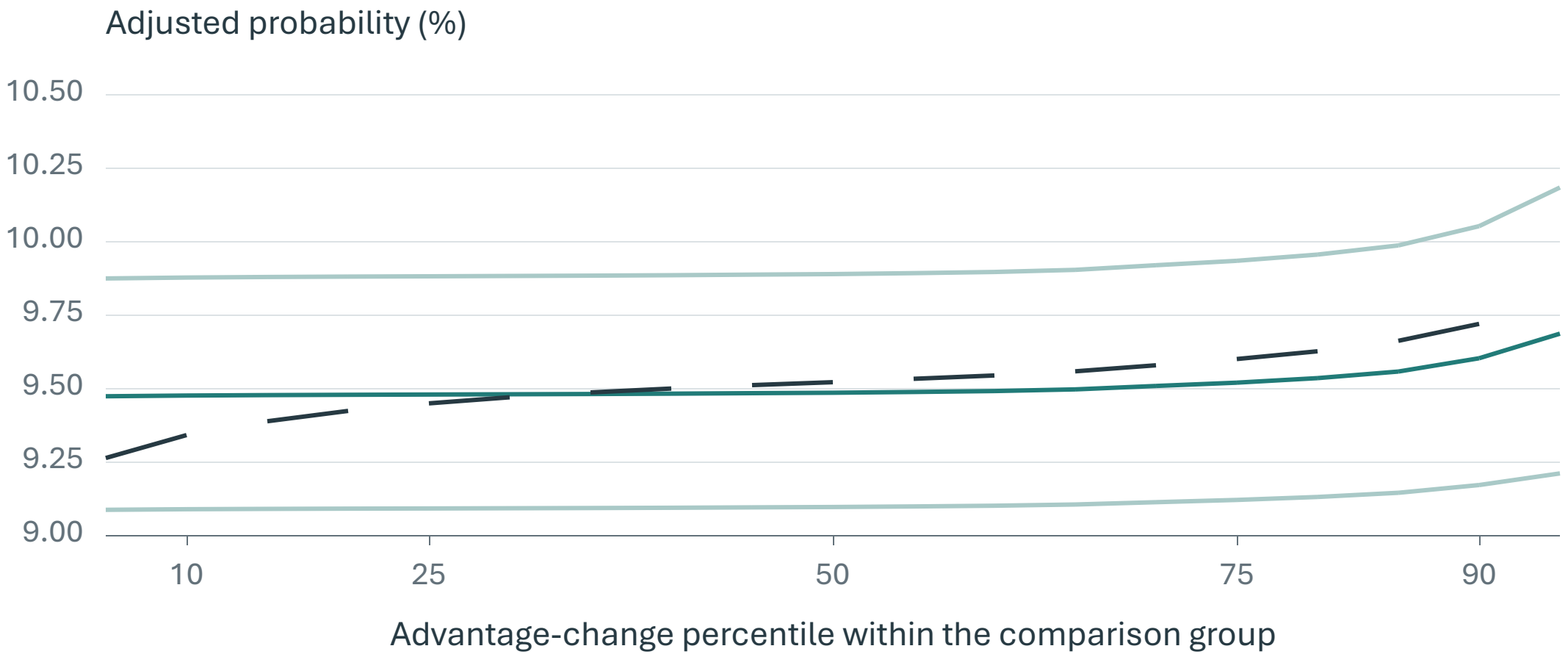
Adjusted Relationship: Incursion Target



Nonlinearity test: p = 0.061

Solid: main logit; dashed: flexible specification. Pointwise 90% intervals.

Adjusted Relationship: Target Selection



Nonlinearity test: p = 0.357

Solid: main logit; dashed: flexible specification. Pointwise 90% intervals.

Expansion, Redistribution and Target Access

A. Unrestricted log-change components

Component or contrast	Origin	Target	Target selection
Common access expansion	-0.069 (0.066)	0.075** (0.032)	-
Relative redistribution	0.081* (0.048)	-0.035 (0.029)	-
Route improvement	-	-	0.030 (0.025)
Target redistribution	-	-	-0.053* (0.029)

Component models retain the original controls, fixed effects and dependence corrections.

Components: Probability Contrasts

B. Substantive contrasts: own-group 25th to 75th percentile

Component or contrast	Origin	Target	Target selection
Common expansion: change in probability (pp)	-0.203 [-0.517, +0.111]	+0.351 [+0.103, +0.600]	-
Redistribution: change in probability (pp)	+0.231 [+0.007, +0.455]	-0.161 [-0.376, +0.054]	-
Route improvement: change in probability (pp)	-	-	+0.027 [-0.009, +0.064]
Target redistribution: change in probability (pp)	-	-	-0.134 [-0.251, -0.017]

Component models retain the original controls, fixed effects and dependence corrections.

Component Tests

C. Diagnostics

Component or contrast	Origin	Target	Target selection
Joint component test: p-value	0.102	0.013	0.013
No common-expansion term: p-value	0.294	0.020	-
Equal/opposite route-target coefficients: p-value	-	-	0.082
Observations	20,547	20,756	28,375
Original controls / fixed effects	Yes	Yes	Yes

Component models retain the original controls, fixed effects and dependence corrections.

Expansion and Redistribution by Firm Role

Firm role	N	Expansion mean (bp)	Redistribution mean (bp)	Redistribution median (bp)
Both	36	16.31	1.376	0.005
Target only	2,402	103.02	0.083	0.008
Origin only	2,123	38.90	0.727	0.022
Origin status unavailable	377	128.60	0.125	0.015
Neither observed	16,164	110.40	0.493	0.025

Log-change components; different from proportional changes in the main descriptive table.

Classification Across Thresholds

A. Classification in the retained firm-invention descriptive sample

Statistic	Elbow	Top 5%	Top 10%
Score cutoff	0.709	0.667	0.617
Retained firm-invention observations	21,102	21,102	21,102
Origin only	2,123	2,568	3,659
Target only	2,402	2,918	3,836
Both roles	36	104	321
Neither observed	16,164	15,149	12,949
Origin status unavailable	377	363	337

Thresholds change labels; the cohort, retention masks and target-selection origins stay fixed.

Classification Across Thresholds: Links

A. Classification in the retained firm-invention descriptive sample

Statistic	Elbow	Top 5%	Top 10%
Observed directed pairs in pair source	2,725	3,577	5,511
Threshold-positive links in event source	4,212	6,008	9,635

Thresholds change labels; the cohort, retention masks and target-selection origins stay fixed.

Estimation Samples Across Thresholds

B. Estimable logistic samples: observations / positive outcomes

Statistic	Elbow	Top 5%	Top 10%
Origin	20,547 / 2,152	20,589 / 2,651	20,646 / 3,933
Target	20,756 / 2,483	20,824 / 3,078	20,906 / 4,234
Directed pair	347,219 / 2,724	347,816 / 3,576	348,821 / 5,510
Target selection	28,375 / 2,701	28,415 / 2,998	28,430 / 3,334

Thresholds change labels; the cohort, retention masks and target-selection origins stay fixed.

Classification Overlap

C. Classification overlap with elbow

Statistic	Elbow	Top 5%	Top 10%
Elbow-positive origins retained as positive	100.0%	100.0%	100.0%

Thresholds change labels; the cohort, retention masks and target-selection origins stay fixed.

Simulation and Target Attribution

D. Elbow simulation and attribution diagnostics

Statistic	Elbow	Top 5%	Top 10%
Positive links with Monte Carlo interval spanning cutoff	1,018	-	-
Positive links with tied leading target	85	-	-
Stable known-origin classifications in retained sample	20,000	-	-

Thresholds change labels; the cohort, retention masks and target-selection origins stay fixed.

Network Associations Across Thresholds

Outcome	Elbow	Top 5%	Top 10%
Incursion origin	1.088* [1.010, 1.174]	1.053 [0.991, 1.119]	1.023 [0.975, 1.073]
Attributed target	0.954 [0.908, 1.003]	0.945** [0.904, 0.989]	0.973 [0.938, 1.009]
Directed incursion	1.009 [0.963, 1.057]	0.990 [0.952, 1.030]	1.021 [0.977, 1.066]
Target selection	1.072* [1.008, 1.141]	1.074* [1.004, 1.150]	1.078** [1.020, 1.140]

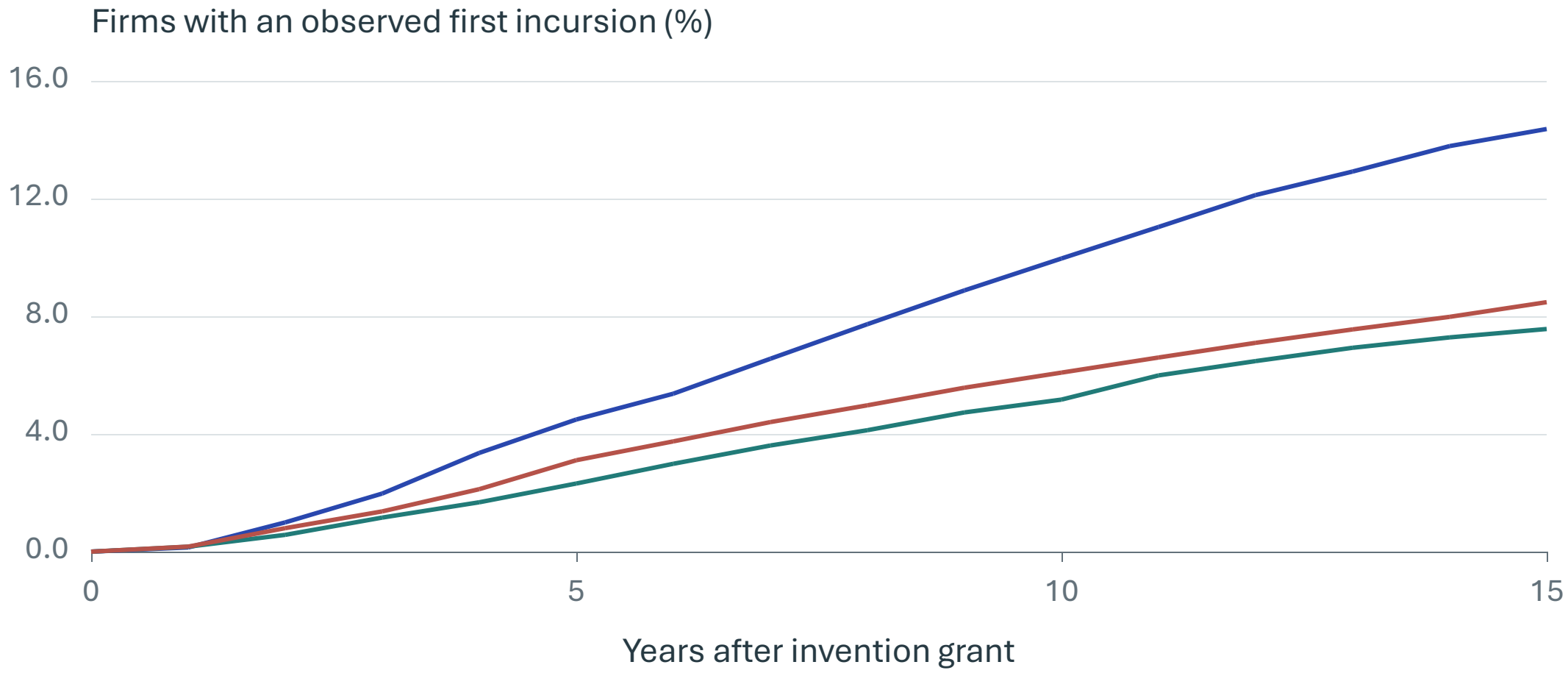
Odds ratios [90% CI], per fixed pooled-design SD. * p < .10; ** p < .05; *** p < .01.

Origin Model: Measurement Sensitivity

Sample	N	Coefficient (SE)	90% coefficient CI
Monte Carlo stable classification (90%)	16,536	0.052 (0.056)	[-0.041, +0.145]
At least ten own priors	4,449	0.071 (0.056)	[-0.022, +0.164]
Numerically resolved balance	19,141	0.086* (0.046)	[+0.010, +0.162]

Same main predictor and controls. These checks change the estimation sample.

When Do Search Incursions Appear?



Middle third: 14.36% **Lower third: 7.57%** **Upper third: 8.48%**

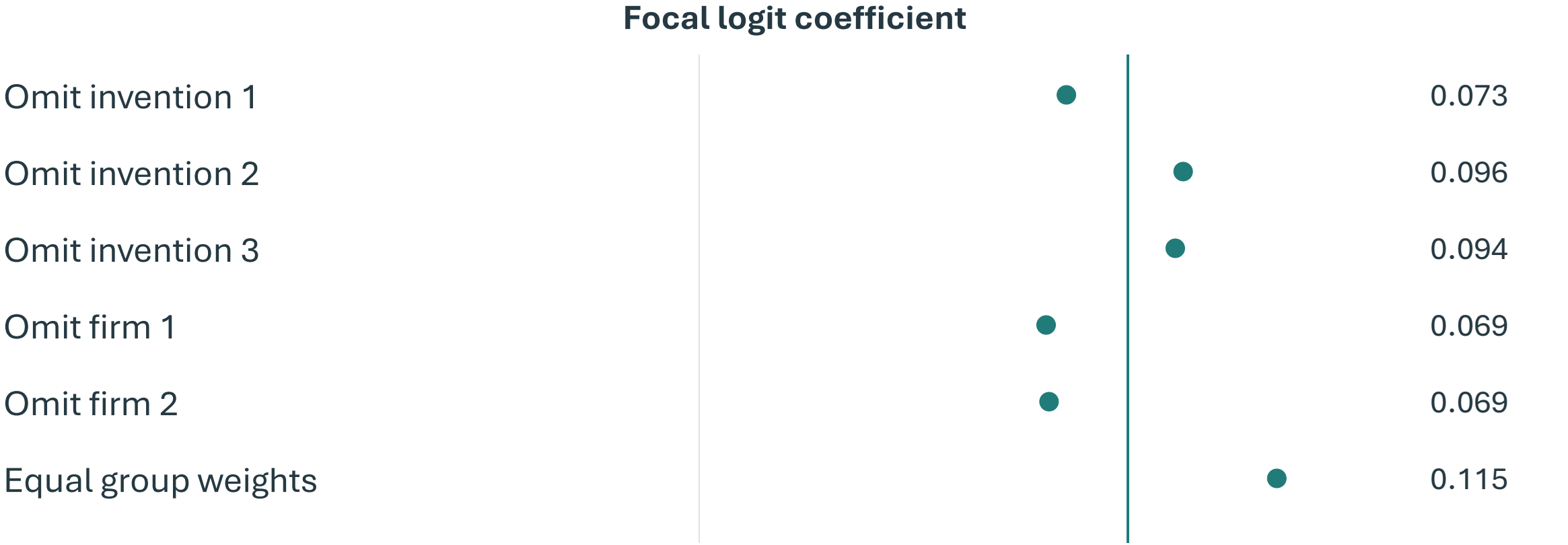
Observed cumulative shares by within-invention redistribution third; not adjusted hazards.

Differences Across Invention Cohorts

Grant cohort	Inventions	Median access expansion (%)	Median redistribution spread (log bp)
1985-1989	229	0.192	1.791
1990-1994	397	0.122	1.815
1995-1999	389	0.247	1.584
2000-2004	523	0.237	1.323
2005-2009	189	0.201	1.096

Spread = within-invention interquartile range of log relative-advantage change.

Influence Checks: Incursion Origin



Original coefficient = 0.085

One influential invention or firm omitted at a time; final row gives equal group weights.

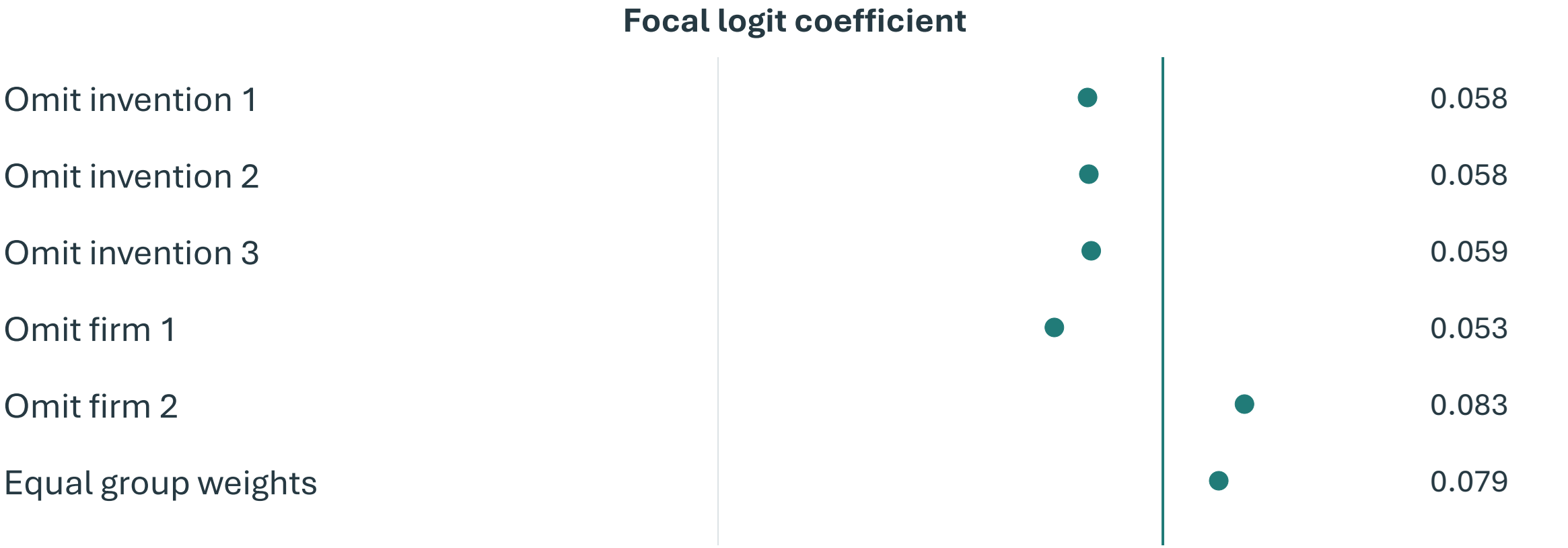
Influence Checks: Incursion Target



Original coefficient = -0.047

One influential invention or firm omitted at a time; final row gives equal group weights.

Influence Checks: Target Selection



Original coefficient = 0.070

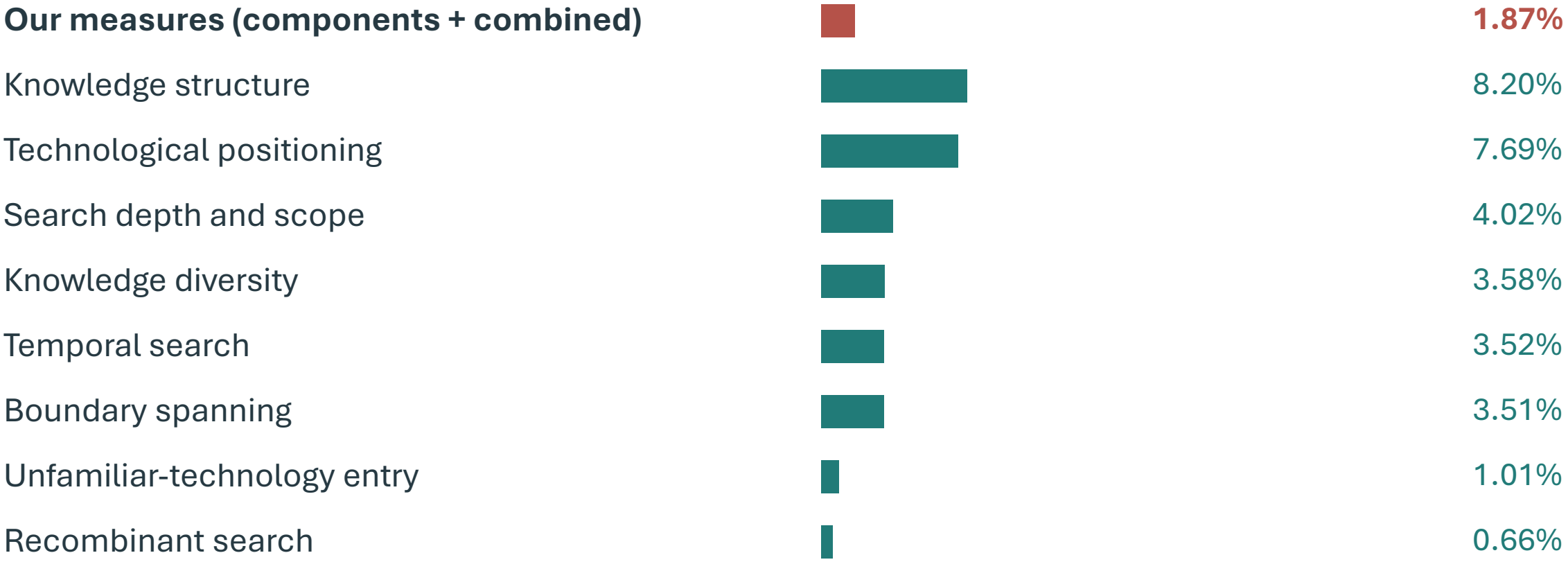
One influential invention or firm omitted at a time; final row gives equal group weights.

Concentration of Network Changes

Measure	N	Median (bp)	Largest 1%: share of total magnitude	Exact zeros
Firm relative advantage	21,102	0.022	10.3%	6.1%
Opportunity	21,102	1.140	16.0%	6.3%
Exposure	21,102	0.301	16.1%	23.2%
Destination-specific advantage	349,353	0.025	29.1%	4.9%
Directed accessibility	349,353	0.000	30.2%	20.4%

Retained sample. Concentration uses absolute change; zeros are exact stored zeros.

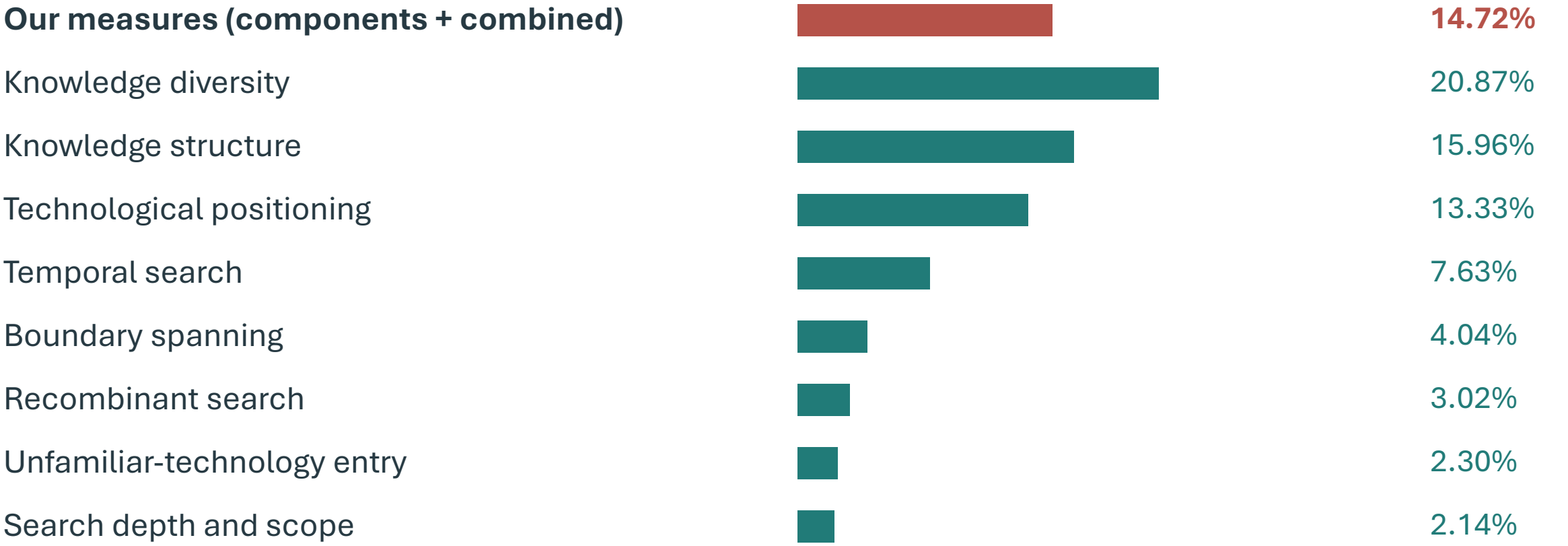
Feature Importance: Incursion Origin



Basic controls: 65.96% of total gain. Our family: 3 predictors.

Native XGBoost total split gain, grouped and normalized. Not an effect or significance test.

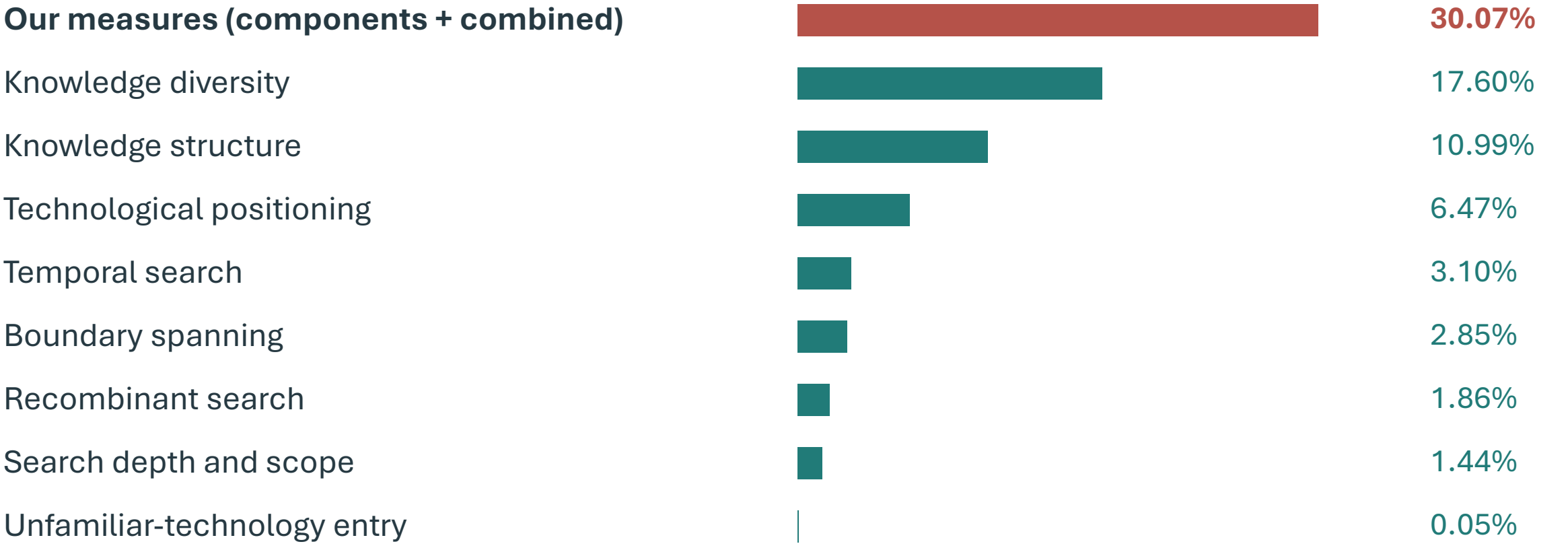
Feature Importance: Incursion Target



Basic controls: 15.99% of total gain. Our family: 3 predictors.

Native XGBoost total split gain, grouped and normalized. Not an effect or significance test.

Feature Importance: Target Selection



Basic controls: 25.57% of total gain. Our family: 8 predictors.

Native XGBoost total split gain, grouped and normalized. Not an effect or significance test.

Top-5 Definition: Full Estimates 1

Predictor	H1 Origin	H2 Target	H3 Target selection
Relative recombinant advantage	0.051 (0.037)	-0.056** (0.027)	-
Destination-specific advantage	-	-	0.072* (0.041)
Prior patent stock (log)	-0.156 (0.129)	-0.418*** (0.084)	-0.369*** (0.087)
Patenting age (log)	-0.158 (0.100)	0.124* (0.069)	0.101 (0.074)
Patenting momentum	0.014 (0.084)	-0.018 (0.057)	-0.043 (0.056)
Prior-patent originality	0.054 (0.047)	-0.023 (0.038)	-0.012 (0.040)
Knowledge-base connectedness	-0.016 (0.048)	-0.012 (0.040)	0.019 (0.043)
One-patent portfolio	0.795*** (0.195)	-0.494** (0.211)	-0.182 (0.206)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-5 Definition: Full Estimates 2

Predictor	H1 Origin	H2 Target	H3 Target selection
Technological breadth	0.144* (0.085)	-0.207*** (0.072)	-0.187*** (0.068)
Distance to invention	0.185*** (0.067)	-0.177*** (0.036)	-0.203*** (0.036)
Selected prior-patent count (log)	-1.318*** (0.140)	0.197** (0.084)	0.096 (0.081)
Search depth (log)	0.322*** (0.096)	-0.147** (0.071)	-0.110 (0.082)
Search scope	-0.321*** (0.098)	-0.094 (0.068)	-0.063 (0.078)
External-citation share	-0.204*** (0.072)	-0.034 (0.043)	-0.013 (0.038)
Knowledge-base vintage	0.115 (0.081)	-0.063 (0.051)	-0.076 (0.052)
External-source nonredundancy	-0.178*** (0.048)	0.082** (0.038)	0.068* (0.038)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-5 Definition: Full Estimates 3

Predictor	H1 Origin	H2 Target	H3 Target selection
Baseline network scale (log)	-1.316*** (0.196)	0.625*** (0.152)	0.567*** (0.179)
Baseline relative advantage (log)	-0.149*** (0.046)	0.038 (0.052)	-0.199 (0.133)
Prior citations: origin to target	-	-	-0.169* (0.092)
Prior citations: target to origin	-	-	-0.155* (0.080)
Baseline directed accessibility	-	-	-0.180 (0.150)
Missing originality	0.251 (0.623)	-0.912 (0.819)	-0.536 (0.790)
Missing breadth	-0.071 (0.336)	0.509* (0.265)	0.383 (0.247)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-5 Definition: Full Estimates 4

Predictor	H1 Origin	H2 Target	H3 Target selection
Missing search depth	-1.372 (0.878)	1.534* (0.895)	0.697 (1.321)
Missing search scope	Omitted	Omitted	Omitted
Missing external-citation share	0.335 (0.525)	0.807 (0.600)	0.970 (0.622)
Missing source nonredundancy	-0.258 (0.308)	-0.632* (0.383)	-0.253 (0.351)
Observations	20,589	20,824	28,415
Positive outcomes	2,651	3,078	2,998
Fixed effects	Invention	Invention	Origin x invention

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-10 Definition: Full Estimates 1

Predictor	H1 Origin	H2 Target	H3 Target selection
Relative recombinant advantage	0.022 (0.029)	-0.028 (0.022)	-
Destination-specific advantage	-	-	0.075** (0.034)
Prior patent stock (log)	-0.030 (0.104)	-0.391*** (0.082)	-0.371*** (0.088)
Patenting age (log)	-0.097 (0.083)	0.112* (0.067)	0.090 (0.071)
Patenting momentum	0.057 (0.065)	-0.030 (0.054)	-0.039 (0.056)
Prior-patent originality	0.058 (0.042)	-0.047 (0.036)	-0.011 (0.040)
Knowledge-base connectedness	0.047 (0.045)	-0.023 (0.037)	-0.007 (0.043)
One-patent portfolio	0.718*** (0.196)	-0.496** (0.221)	-0.272 (0.223)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-10 Definition: Full Estimates 2

Predictor	H1 Origin	H2 Target	H3 Target selection
Technological breadth	0.170** (0.073)	-0.186*** (0.069)	-0.209*** (0.068)
Distance to invention	0.141** (0.055)	-0.180*** (0.032)	-0.192*** (0.037)
Selected prior-patent count (log)	-1.133*** (0.117)	0.118 (0.081)	0.068 (0.081)
Search depth (log)	0.223*** (0.076)	-0.146** (0.072)	-0.120 (0.086)
Search scope	-0.301*** (0.084)	-0.117* (0.064)	-0.102 (0.076)
External-citation share	-0.083 (0.052)	-0.013 (0.043)	-0.002 (0.038)
Knowledge-base vintage	0.103 (0.070)	-0.058 (0.049)	-0.050 (0.049)
External-source nonredundancy	-0.167*** (0.041)	0.076** (0.038)	0.070* (0.037)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-10 Definition: Full Estimates 3

Predictor	H1 Origin	H2 Target	H3 Target selection
Baseline network scale (log)	-1.265*** (0.164)	0.635*** (0.137)	0.582*** (0.173)
Baseline relative advantage (log)	-0.154*** (0.044)	0.046 (0.045)	-0.136 (0.125)
Prior citations: origin to target	-	-	-0.176** (0.087)
Prior citations: target to origin	-	-	-0.097 (0.084)
Baseline directed accessibility	-	-	-0.150 (0.142)
Missing originality	0.218 (0.633)	-0.205 (0.683)	-0.373 (0.756)
Missing breadth	-0.277 (0.306)	0.465* (0.259)	0.399 (0.248)

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Top-10 Definition: Full Estimates 4

Predictor	H1 Origin	H2 Target	H3 Target selection
Missing search depth	Omitted	Omitted	Omitted
Missing search scope	-1.435* (0.802)	0.642 (0.855)	0.258 (1.197)
Missing external-citation share	0.404 (0.512)	0.198 (0.544)	0.818 (0.629)
Missing source nonredundancy	-0.241 (0.292)	-0.400 (0.322)	-0.299 (0.361)
Observations	20,646	20,906	28,430
Positive outcomes	3,933	4,234	3,334
Fixed effects	Invention	Invention	Origin x invention

Logit coefficients (SE). * p < .10; ** p < .05; *** p < .01.

Measures Used in the Models

Directed accessibility	a(ij,s) = 1 / [1 + C(ij,s)]
Recombinant opportunity	O(i,s) = sum of a(ij,s) over other firms j
Recombinant exposure	E(i,s) = sum of a(ji,s) over other firms j
Relative recombinant advantage	R(i,s) = O(i,s) / E(i,s)
Destination-specific advantage	S(ij,s) = a(ij,s) x R(i,s) / R(j,s)
Invention-induced change	After / before - 1

s = before or after the invention. Changes compare after with before.

How to Read the Appendix

Coefficients	Log-odds estimates. Parentheses contain standard errors.
Probabilities	Model-adjusted averages at within-group predictor percentiles.
Uncertainty	90% intervals. Stars: * p < .10, ** p < .05, *** p < .01.
Sample	1985-2009 inventions; corporate adoption in the following 15 years.
Feature importance	Share of XGBoost split gain, not a causal effect or significance.
Provenance	Sources and construction details are retained in the project records.

All results reproduce the current saved estimates. No new estimation.